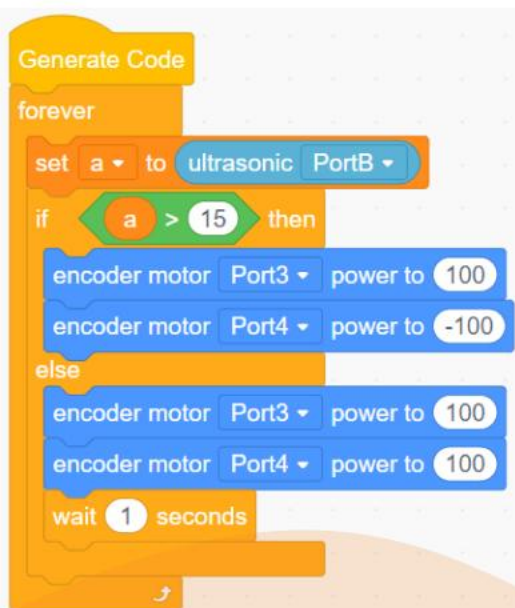


12 in 1 WeeeBot RobotStorm V2.0

WeeeCode Program Activities Book



```
#include <WeELF328P.h>
```

```
WeUltrasonicSensor ultrasonic_B(PORT_B);
WeEncoderMotor encoder_3(PORT_3);
WeEncoderMotor encoder_4(PORT_4);
double _a = 0; //a;
```

```
void setup() {
}
```

```
void loop() {
  _a = ultrasonic_B.distanceCm();
  if(_a > 15){
    encoder_3.run(100);
    encoder_4.run(-100);
  }else{
    encoder_3.run(100);
    encoder_4.run(100);
  }
}
```



A. Walking robot

Material

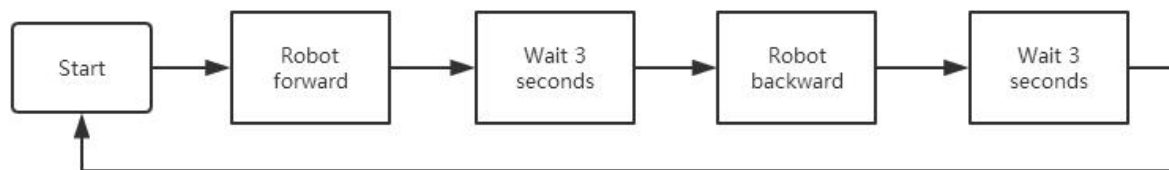
1. Walking robot
2. Computer



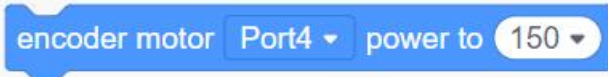
Project #1 Control Encoder DC Motor

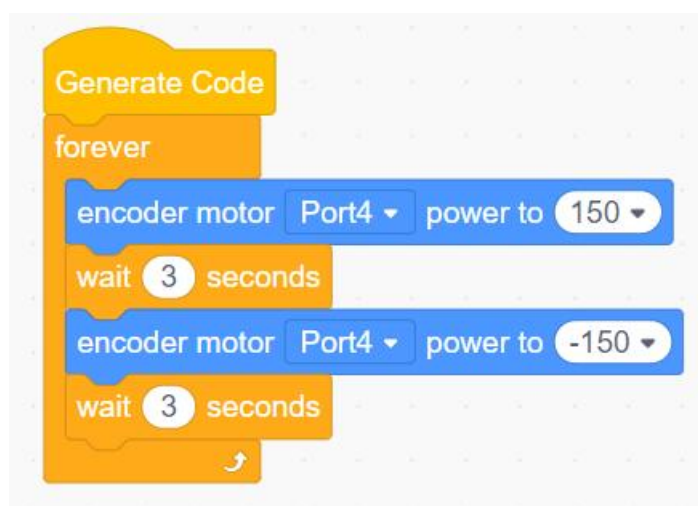
Target: Make the walking robot go forward for 3 seconds and backward for 3 seconds.

Program Idea



Reference Code

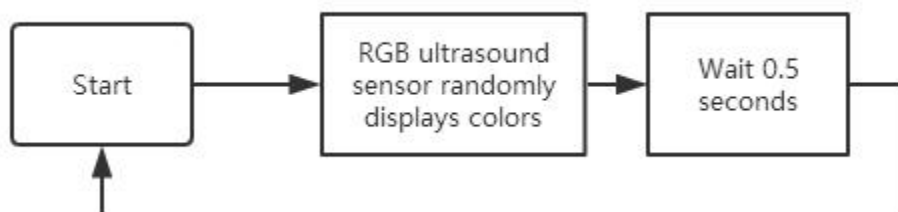
	Generate Arduino code of the program below, the Arduino code can be check on “code” window, and upload to board.
	The Forever block is a Control block. Blocks held inside this block will be repeated and the loop never ends (unless the stop sign is clicked, the Stop All block is activated, or the stop script block is activated within the loop).
	This block pauses its script for the specified amount of seconds the wait can also be a decimal number, it is used whenever a sprite must wait for another action.
	This block defines the speed and rotation direction of encoder DC motor. Range from -255 to 255.



Project #2: Control RGB LED

Target: Control RGB LED to randomly display one color every 0.5 seconds.

Program Idea



Reference Code

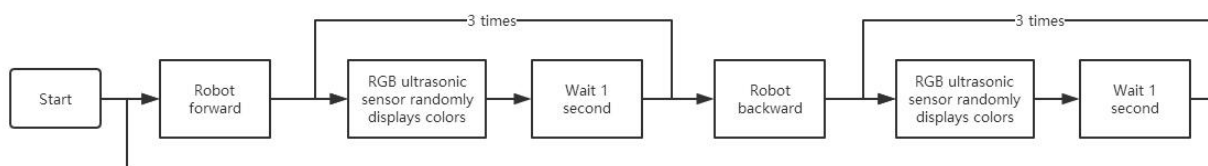
	This block defines the color of two RGB LEDs on ultrasonic sensor. Select RGB LED you want to control and set RGB values. R refers to red [0,255], G refers to green [0,255], B refers to blue [0,255].
	This block picks a pseudorandom number ranging from the first given number to the second, including both endpoints. If both numbers have no decimals, it will report a whole number.



Project #3: Combine Encoder DC Motor with RGB LED

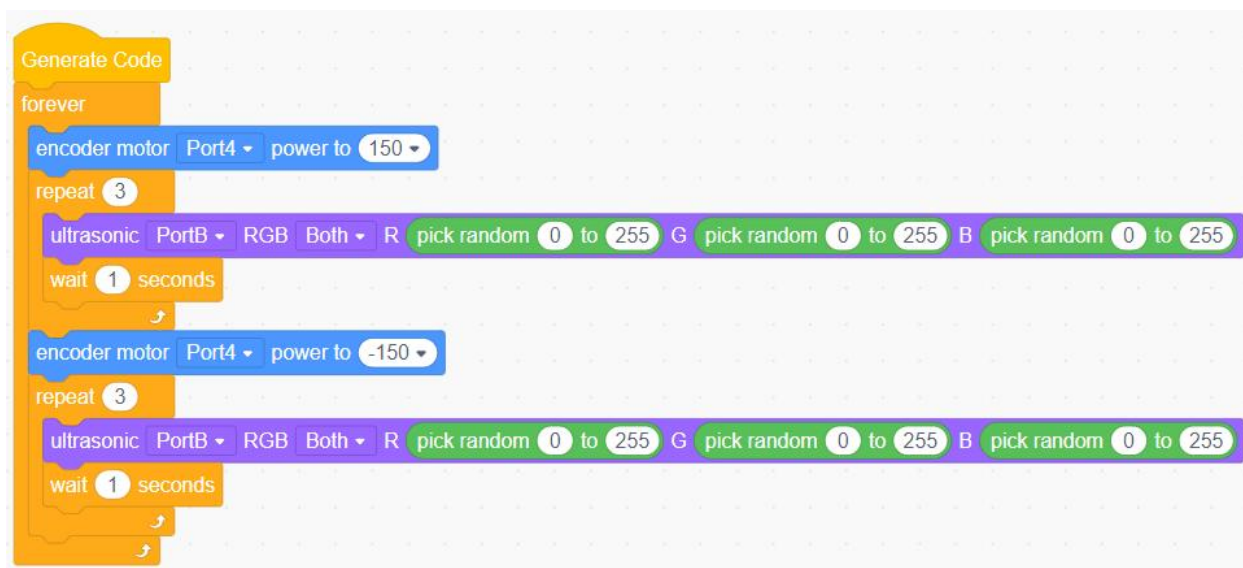
Based on the first project, we modified the block [wait (3) secs] to block [repeat (3)]. During the repeated execution, the RGB ultrasonic sensor can change colors, with an interval of 1 second for each color change. This is consistent with the time we waited for 3 seconds before, and the robot can change colors while walking.

Program Idea



Reference Code

	Blocks held inside this block will loop a given amount of times, before allowing the script to continue. If a decimal is put in, the number is rounded up. Furthermore, when a non-positive number is input, the loop does not run, and if "Infinity" is input, then the block runs forever.
---	---



Expansion: You can try to program and control the robot's walking rhythm and light change to correspond to the song rhythm, so that the robot can dance with the music rhythm.

B. Rolling Tank

Material

1. Rolling Tank
2. Computer

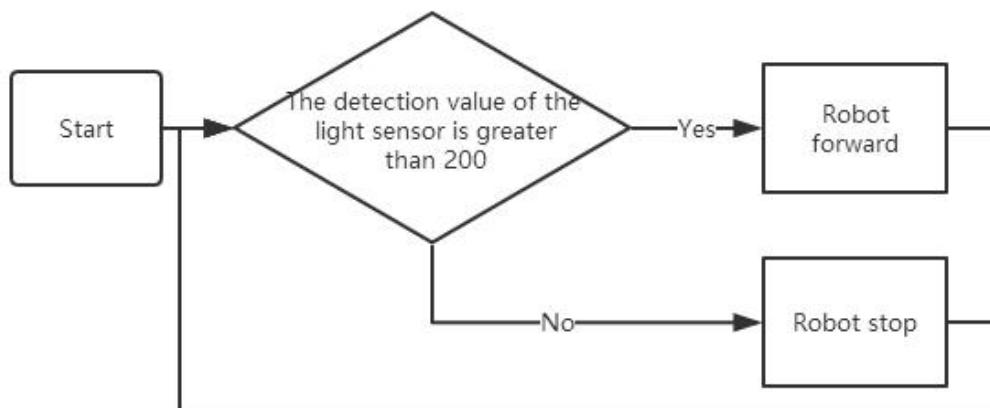


Project #1 Light-driven Robot

Task: Make a robot moving per the light value in environment.

The light intensity in the environment is detected by the light sensor. If in normal environment, the light detected value is 300, and the value change to 100 while covering the light sensor with hand, we can take the mid-value 200 as the judgment value. When the value is greater than 200, the robot would go forward. When the value is less than 200, the robot stops.

Program Idea

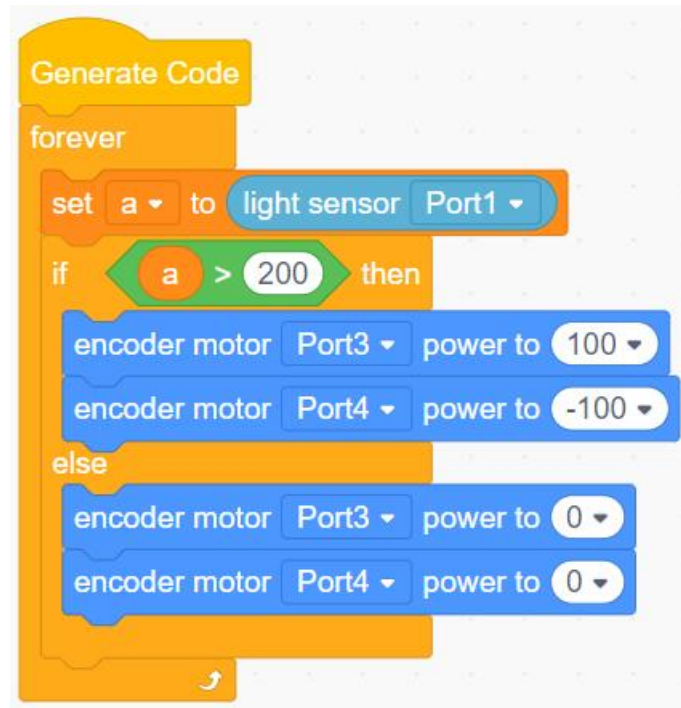


Reference Code

	The block will check its boolean condition. If the condition is true, the code held inside the first C (space) will activate, and then the script will continue; if the condition is false, the code inside the second C will activate
	The block reports the value of light intensity detected by light sensor.

Make a Variable

	The block simply holds its variable. Whenever a variable is created, a version of the block appears with the variable's given name on it this results in a version of this block for every variable. Each version of the block holds its 'assigned' variable.
	The block defines the specified variable to the given value: a string or number.
	The block will change the specified variable by a given amount. If the variable is a string and not a number, it is set to the quantity the variable was to be changed by (casting the string to 0).
	The block shows the specified Variable's Stage monitor.
	The block hides the specified variable's Stage monitor.

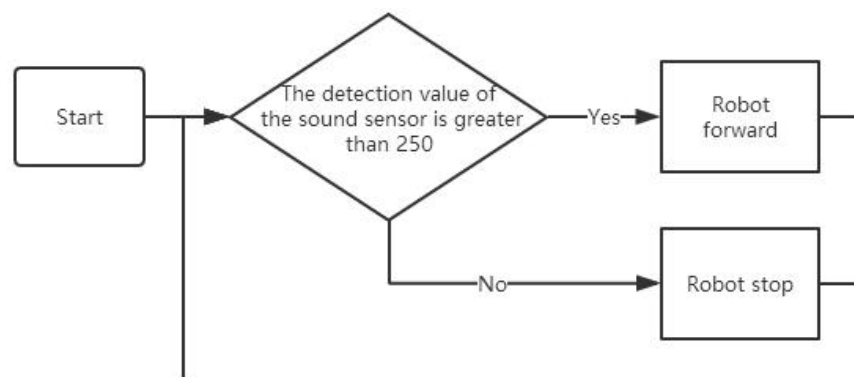


Project #2: Sound-driven Robot

Task: Make a robot moving per the sound value in environment.

The sound intensity in the environment is detected by the sound sensor. If in normal environment, the sound detected value is 100, and then change to 400 while the environment is noisy, we can take the mid-value 250 as the judgment value. When the value is greater than 250, the robot would go forward. When the value is less than 250, the robot stops.

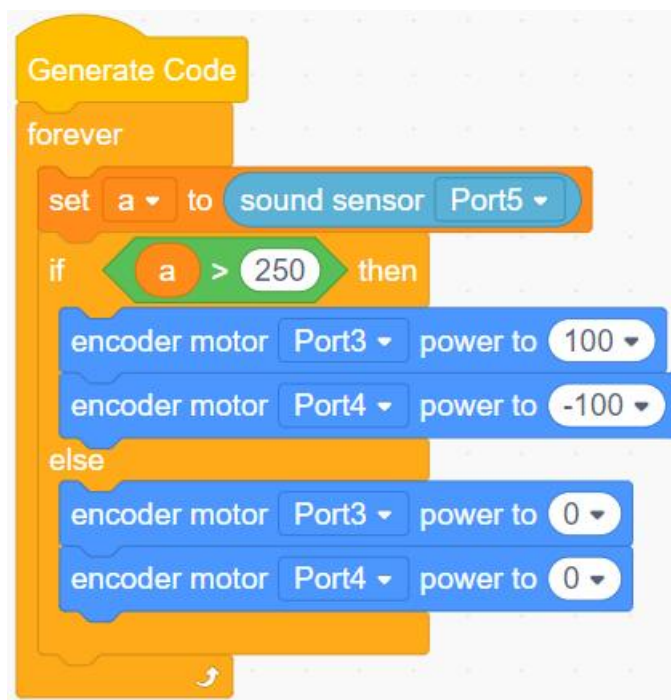
Program Idea



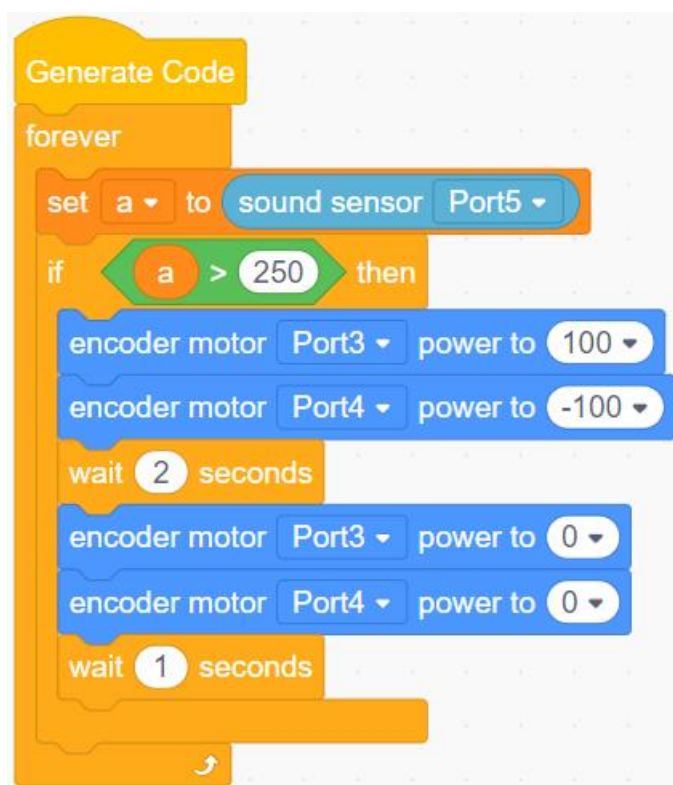
Reference Code 1

sound sensor Port3 ▾

The block reports the value of volume detected by sound sensor.



Reference Code 2



Note: Before re-judging the detection of the sound sensor, the robot should be allowed to stop firstly, to avoid the noise during the movement of the robot interferes with the detected value of the sound sensor.

Expansion: You can try to make a "coward" robot by combining light sensor with sound sensor, it will only move when there is no one at night.

C. Spider Robot

Material

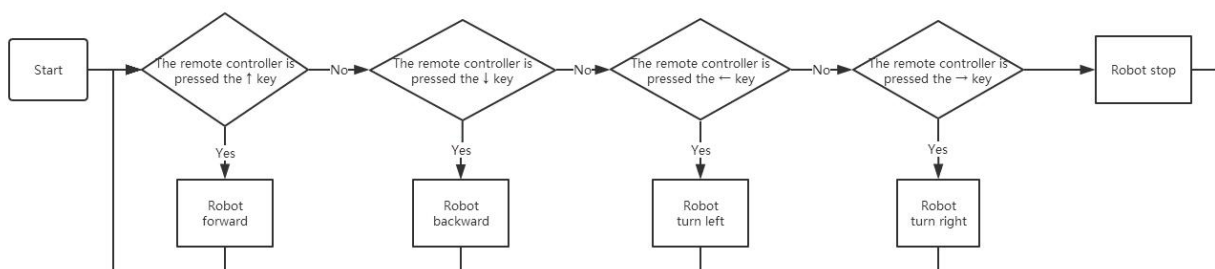
1. Spider Robot
2. Computer



Project #1: Infrared Remote Controller Controls Robot

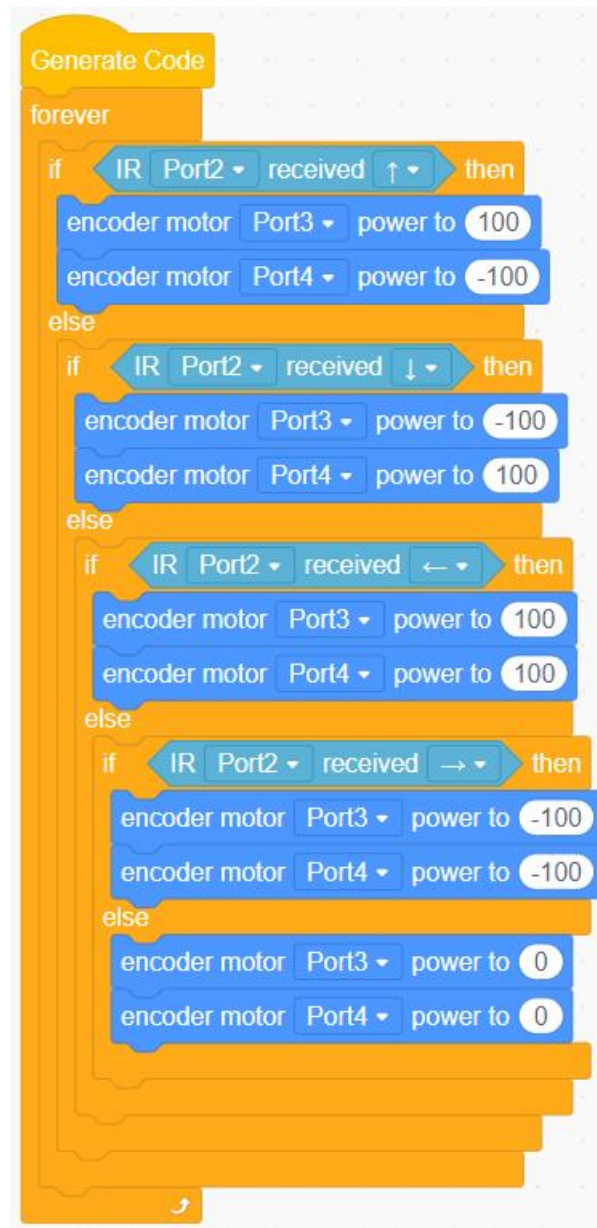
Use the IR remote controller to control the movement of the robot: if IR remote controller is pressed the ↑ key, the robot goes forward; if IR remote controller is pressed the ↓ key, the robot goes back; if IR remote controller is pressed the ← key, the robot turns left; if IR remote controller is pressed the → key, the robot turns right.

Program Idea



Reference Code

	The block checks if the specified key of IR remote controller is pressed. If the key is being pressed, the block returns "true"; if it is not, it returns "false".
---	---



Project #2: Singing Robot

Task: Control the buzzer module to broadcast the sound according to the notes and tones of the music.

Program Idea

Little Star

$$1=C \frac{4}{4}$$

1 1 5 5 | 6 6 5 - | 4 4 3 3 | 2 2 1 - | 5 5 4 4 | 3 3 2 - |

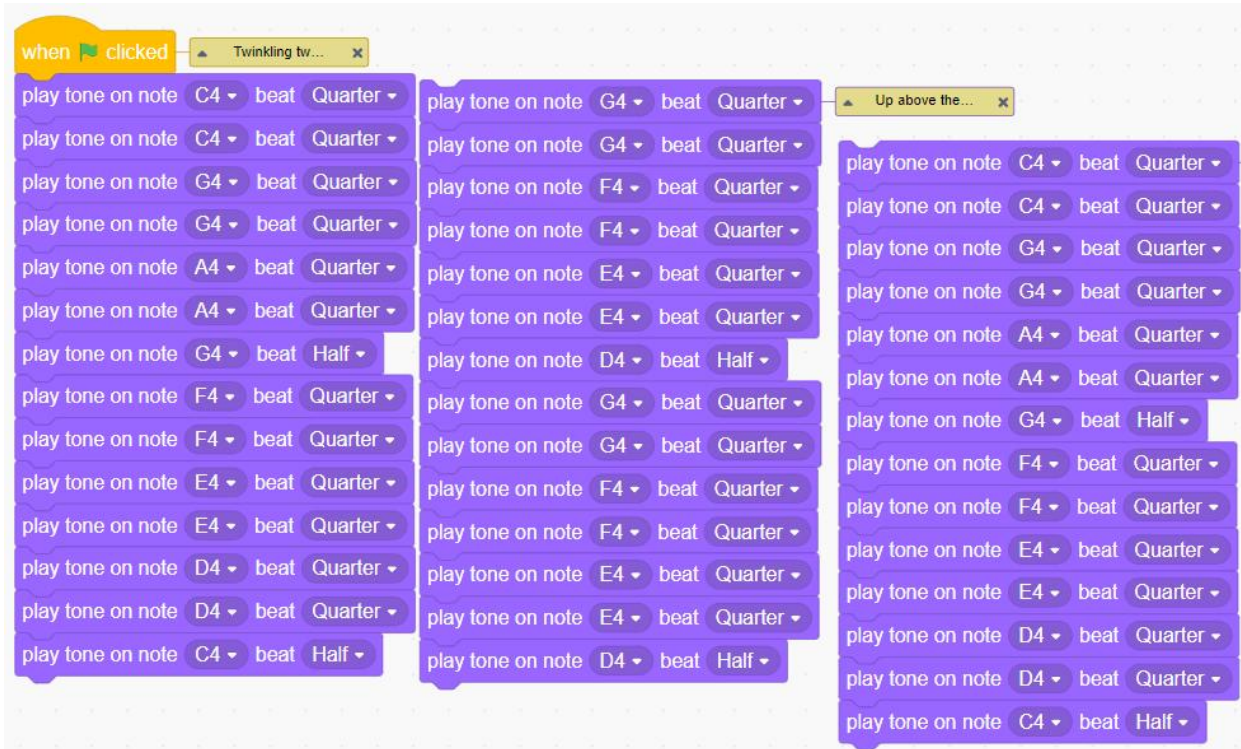
Twinkle twinkle ,little star, How I wonder what you are, Up above the world so high,

5 5 4 4 | 3 3 2 - | 1 1 5 5 | 6 6 5 - | 4 4 3 3 | 2 2 1 - |

Like a diamond in the sky. Twinkle twinkle,little star, How I wonder what you are.

Reference Code

	Scripts that wear this block will activate once the Green Flag has been clicked these scripts can activate other scripts and enable the entire program.
	This block defines the tone and rhythms of the buzzer.



Expansion: You can make a music robot by pressing different buttons of the remote controller, and the robot sings different songs.

D. Detection Robot

Material

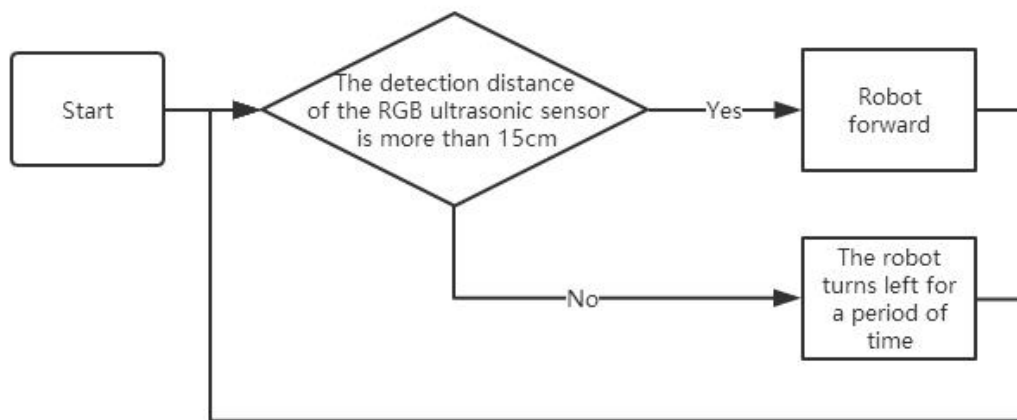
1. Detection Robot
2. Computer



Project #1: Obstacle Avoidance Robot

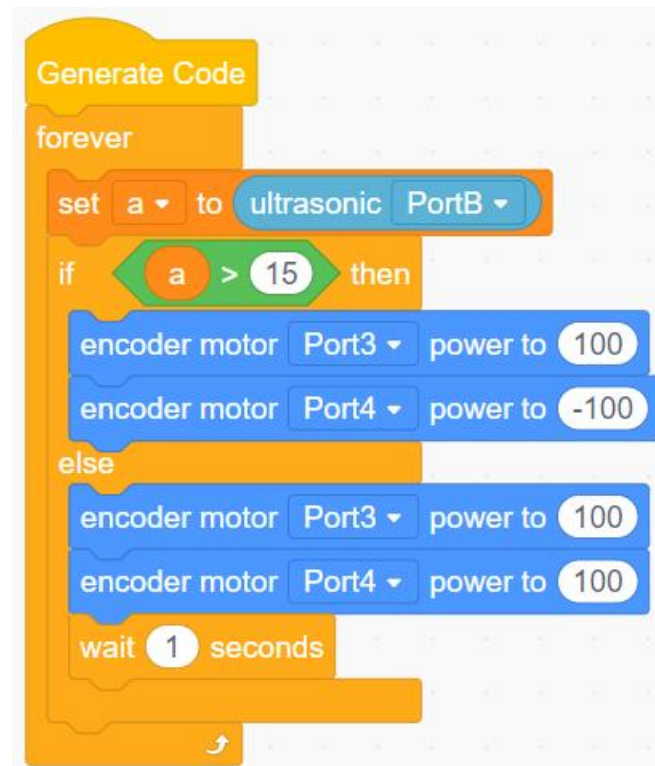
Task: Use RGB ultrasonic sensor to detect whether there is an obstacle in front of the robot. If RGB ultrasonic sensor detects a small value, it means there is an obstacle in front, and the robot will turn. If RGB ultrasonic sensor detects a large, it means there is no obstacle ahead and the robot will move forward.

Program Idea



Reference Code

	The block reports the distance between the ultrasonic sensor and object in front of it in center meter (cm).
	The block checks if the first value is greater than the other value. If the second value is less, the block returns true; if not, it returns false. This block works with letters too, not just numbers. Letters at the top of the alphabet (e.g. a, b, c) are worth less than letters at the end (e.g. x, y, z). Other rules of alphabetical order apply (e.g. zach > az).

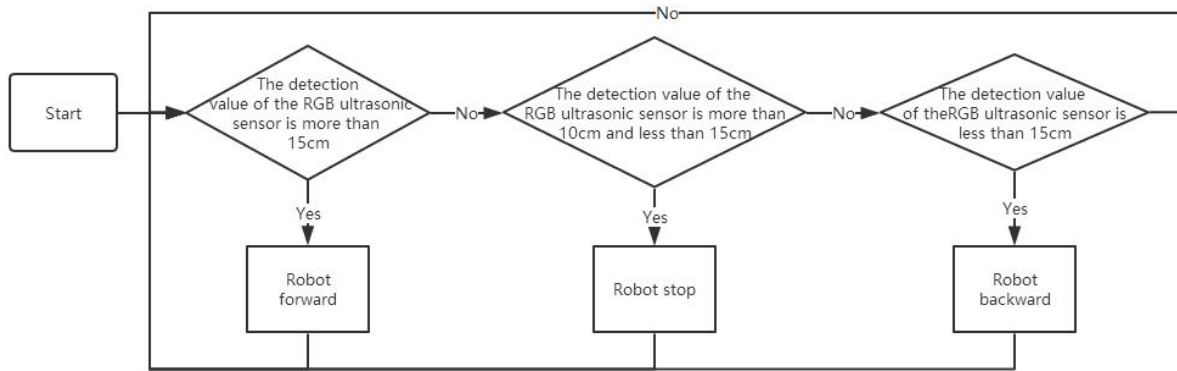


Project #2: Following Robot

Task: Make a following robot.

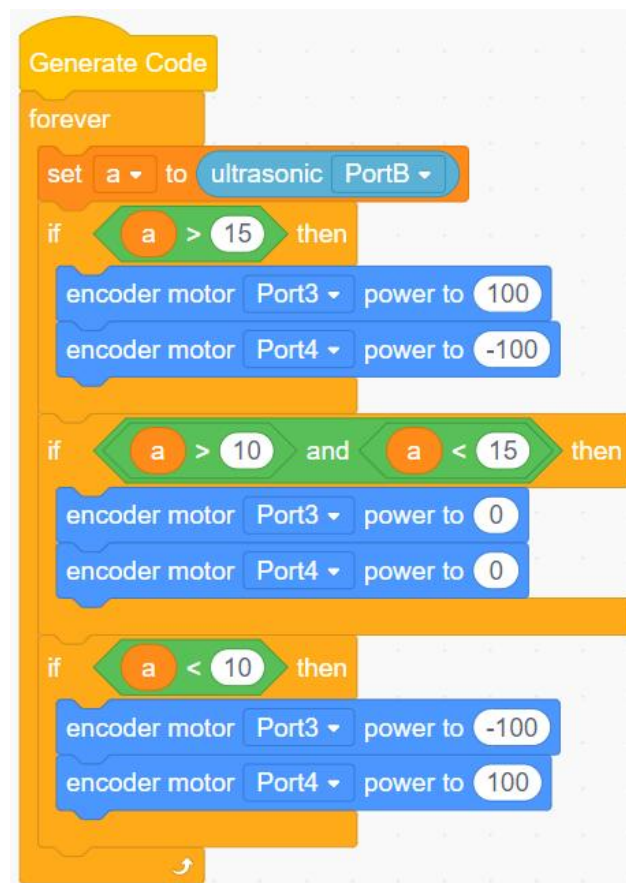
The robot changes its motion state according to the detected distance of the RGB ultrasonic sensor. When the RGB ultrasonic sensor detects a small value, it means the robot is close to the object in front, then the robot moves back. When the detected value remains unchanged within an interval, then the front object can be considered stationary, and the robot stop. When the RGB ultrasonic sensor detects a large value, it means the robot is far away from the front object, then the robot moves forward. In this way, the effect of the robot following the movement of the object is achieved.

Program Idea



Reference Code

	The block checks if the first value is less than the second value. If it is less, the block returns true; if not, it returns false.
	The block joins two boolean blocks so they both have to be true to return true. If they are both true, the block returns true; if they are not all true or none true, it returns false.





Note: The RGB ultrasonic sensor detected value, which can make robot stop moving, must be an interval. Because RGB ultrasonic sensor detected value will wave, if it is not a range but a specific value, the robot will move back and forward instead of being static.

Expansion: You can make a more intelligent robot: when there are obstacles in front of the robot, the robot don't go forward even if the remote controller buttons were pressed, and the RGB light will turn on the red light, the buzzer module will sound an alarm to remind the operator that there are obstacles in front of the robot, the robot can't go forward.

E. 3-wheel Robot

Material

1. Three-wheel Robot
2. Computer



Project #1: Line-following Robot

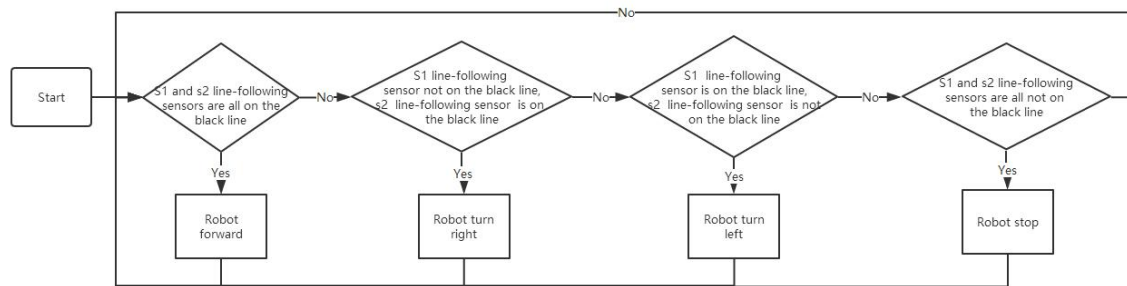
Task: Make a line-following robot.

There are two ways of robot line-following: ① the black line clip line-following sensor (the line is wider than the sensor spacing); ② the line-following sensor clip the black line (the sensor spacing is wider than the line).

- ① The black line clips the line inspection sensor (the line width is larger than the sensor spacing).



Program Idea



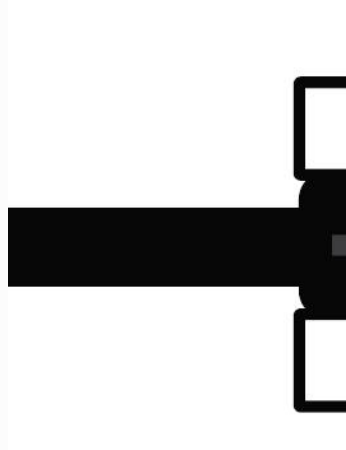
Reference Code

line follower PortA ▼ S1 ▼	The block reports the reflection value of S1 and S2 (0-1023), the value will be a significant difference between black color and white color.
---	--

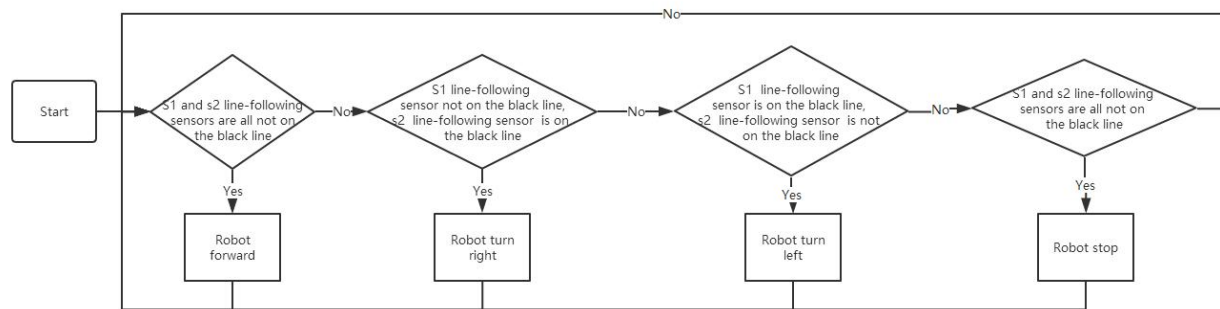


Note: The line-following inspection judgment value needs to be adjusted according to the actual line-following inspection environment.

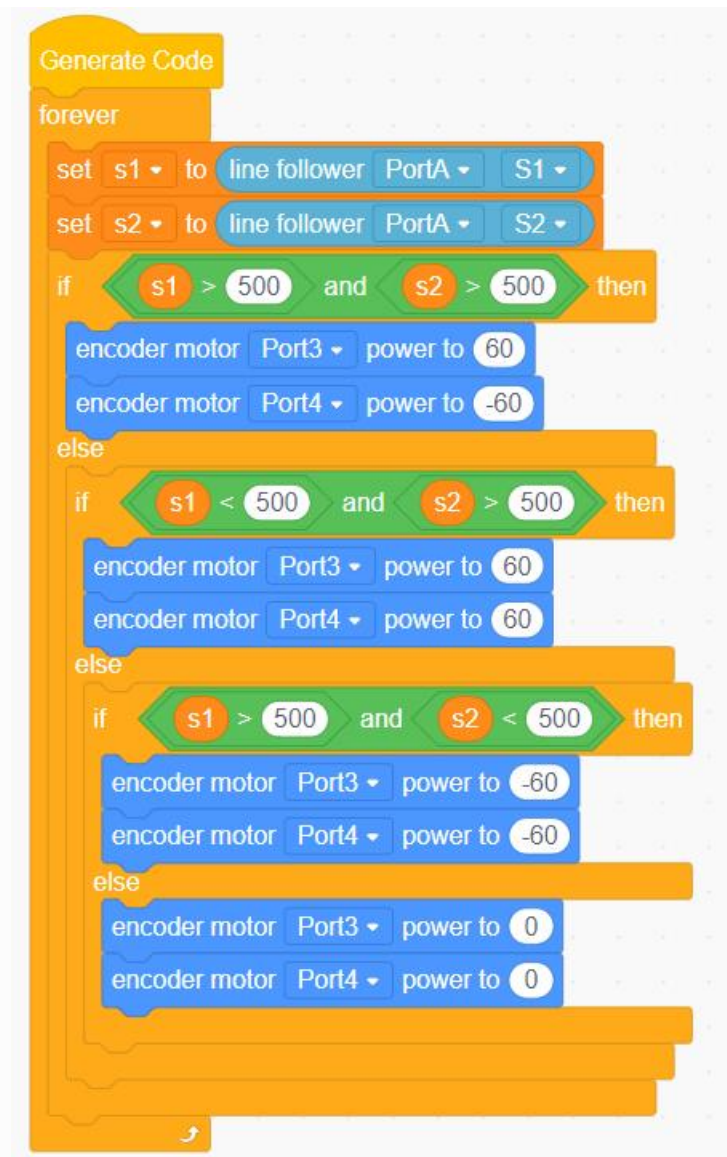
- ② The line-following sensor clips the black line (the sensor spacing is wider than the line).



Program Idea



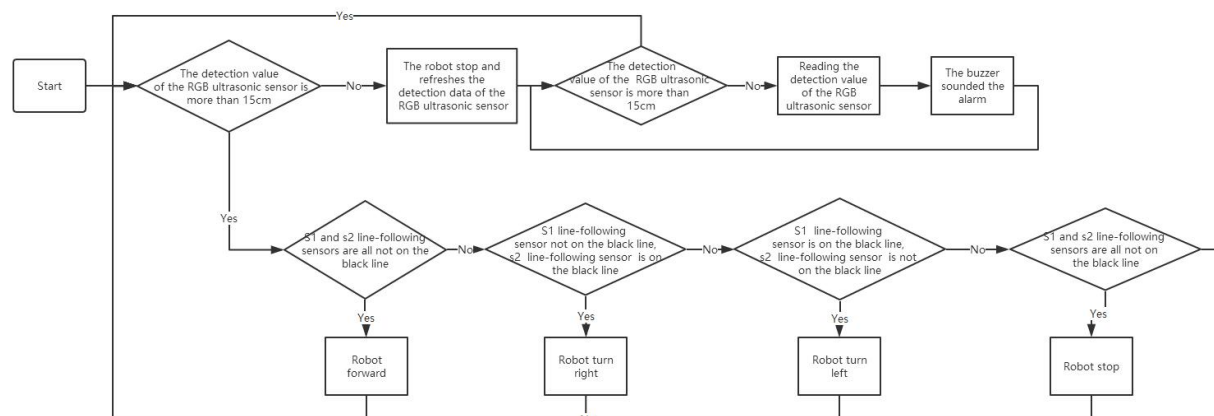
Reference Code



Project #2: Combine Line-following with Obstacle Avoidance

Like the feeding robot, when the robot encounters an obstacle in front of the line, the robot stops line-following and issues an alarm. If there is no obstacle in front, the robot will do line-following normally.

Program Idea





Reference Code

forever

```
if (Detection value of ultrasonic sensor > 15) then
```

```
set s2 to line follower PortA S2
```

encoder motor Port3 ▾ power to 60

else

encoder motor Port3 ▾ power to 60

else

encoder motor Port3 ▾ power to -60

else

encoder motor Port4 ▾ power to 0

encoder motor Port4 ▾ power to 0

```
repeat until < Detection value of ultrasonic sensor > 15
```

play tone on note C4 ▾ beat Half ▾



Expansion: Try to design more complex line-following route and use more optimized line-following algorithms or more line-following sensors to enable robots to work.

F. Forklift Robot

Material

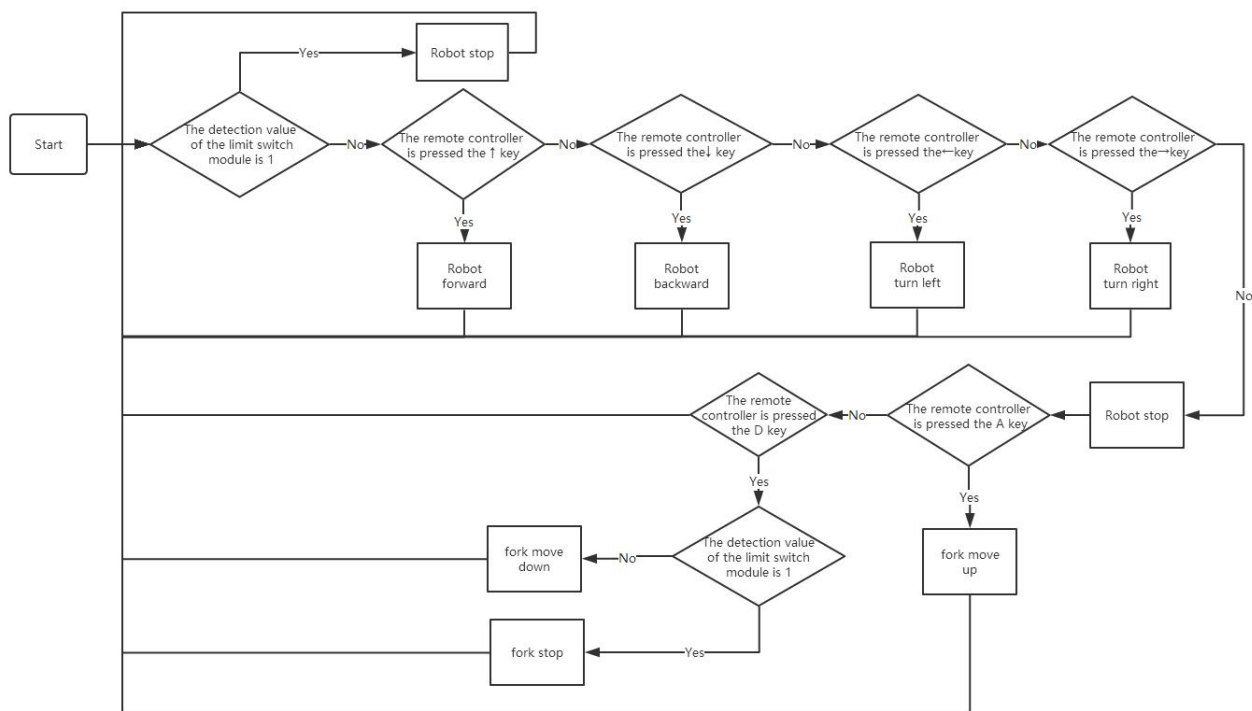
1. Forklift Robot
2. Computer



Project #1: IR Control Robot with Limit Switch

Use the up, down, left and right buttons of the remote controller to make robot move forward, backward, left turn and right turn, use the A and D buttons of the remote controller to control up and down movement of fork of the forklift robot, and limit the moving range of the fork through the limit switch module.

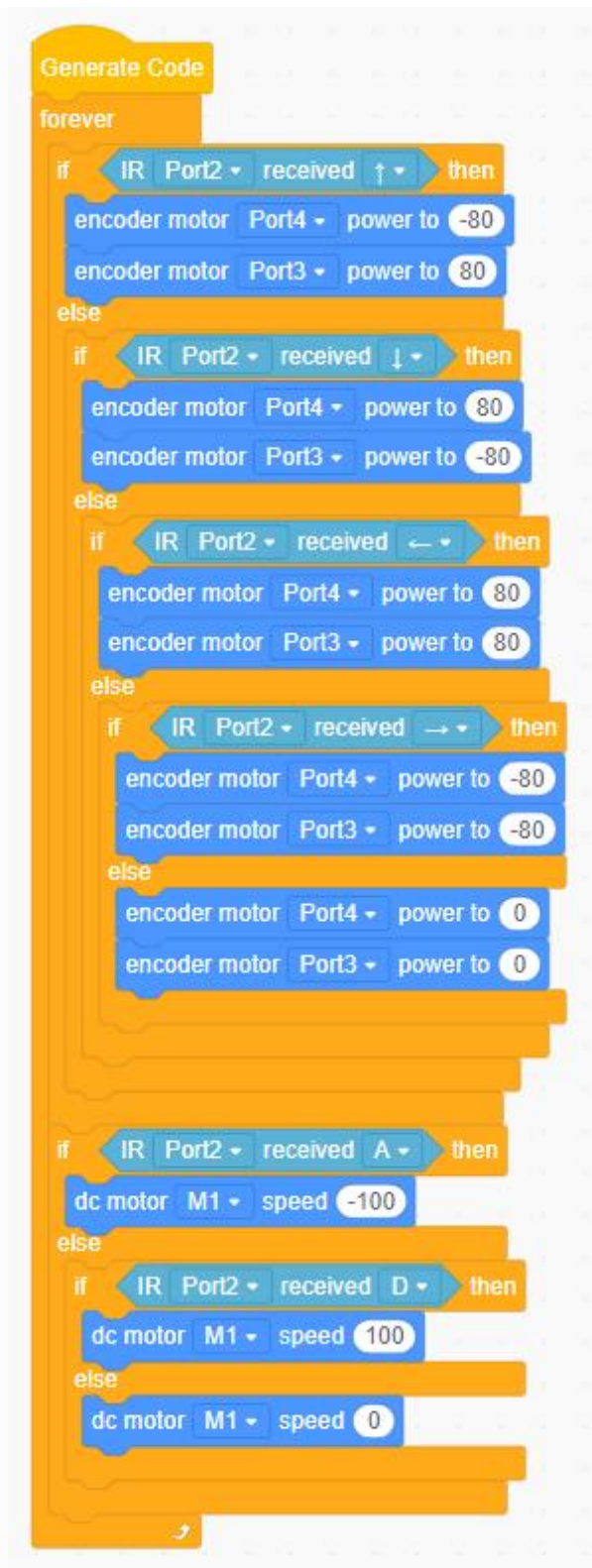
Program Idea



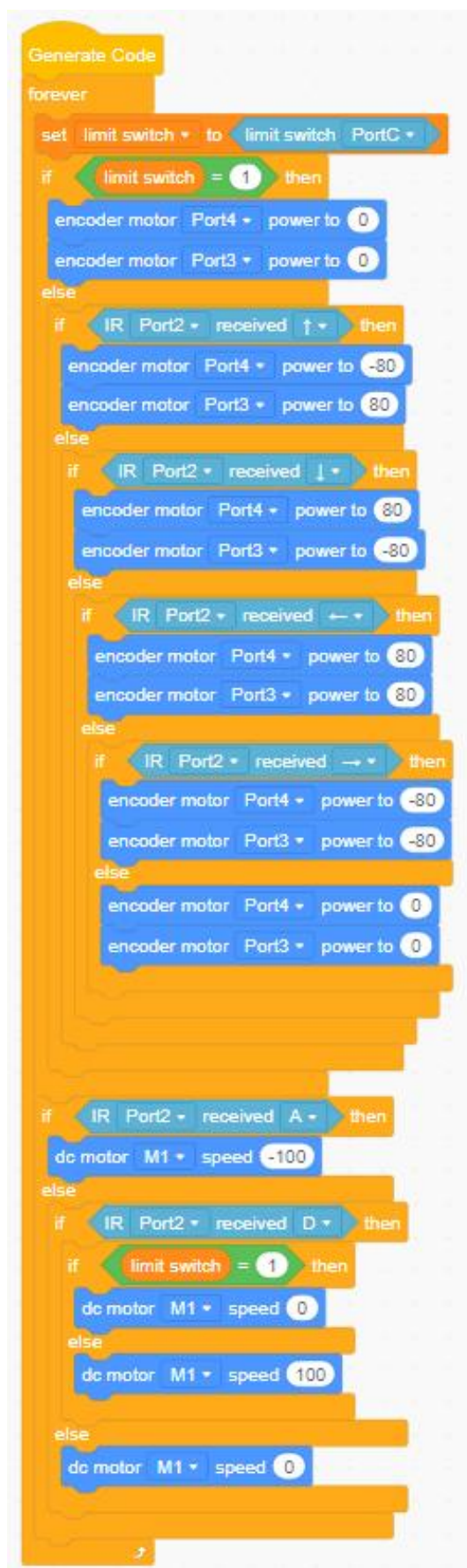
Note: When the detected value of limit switch module is 1, press the direction key of the remote controller, the robot cannot move, otherwise the limit switch module will be damaged.

Reference Code – IR Control Robot

	The block checks if the plunger of limit switch is pressed. If the plunger is being pressed, the block returns "true"; if it is not, it returns "false".
--	---



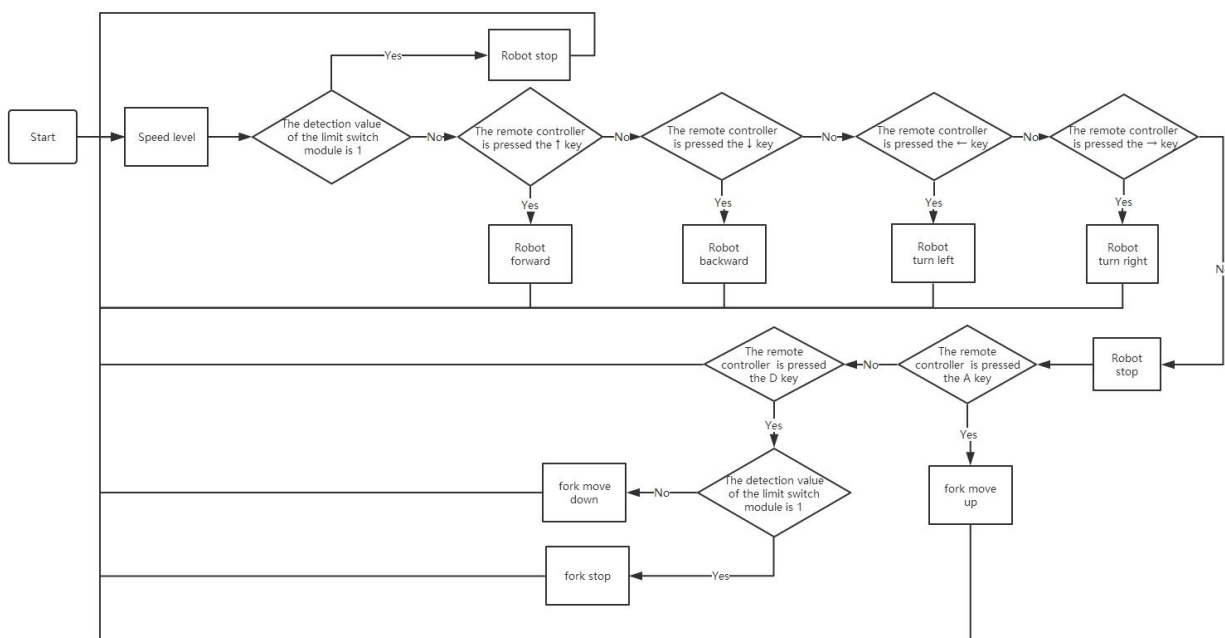
Reference Code – IR Control Robot with Limit Switch

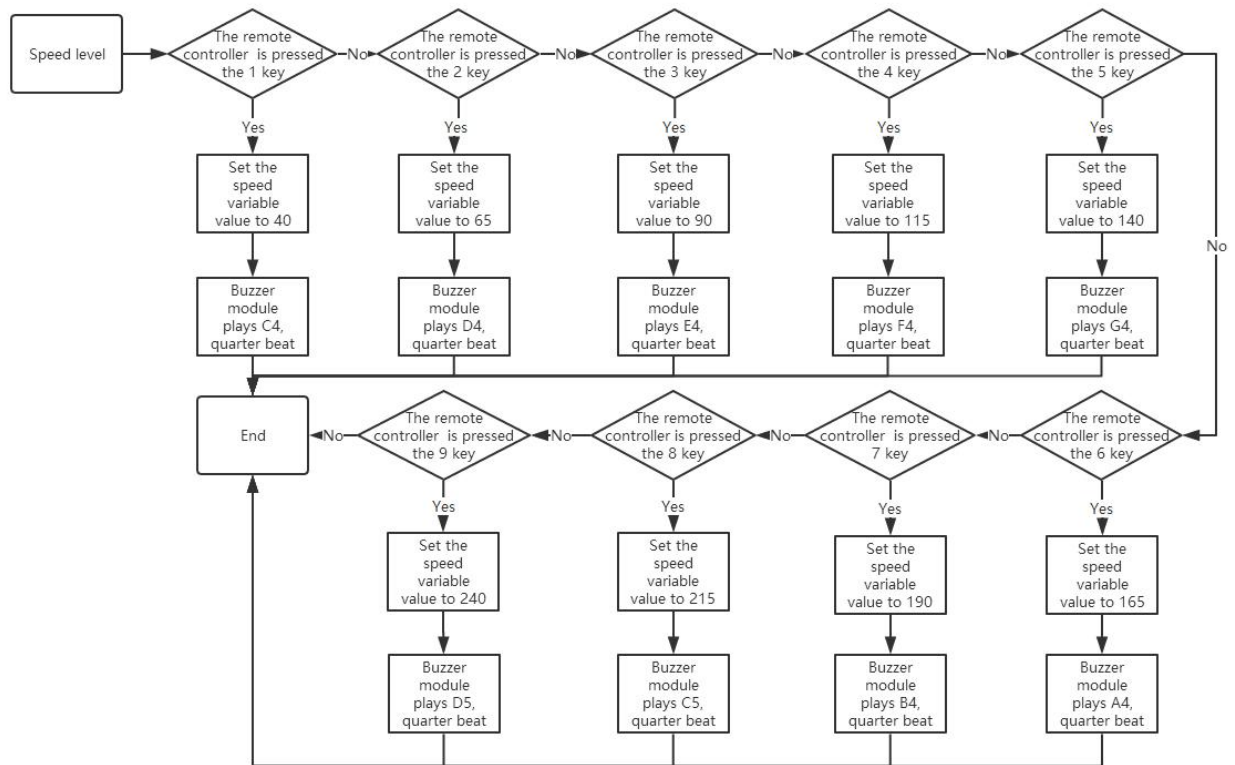


Project #2: IR Speed Control Forklift Robot

Use the up, down, left and right buttons of the IR remote controller to control the movement of the robot in all directions, use the A and D buttons of the remote controller to control the up and down movement of the fork of the forklift robot, use the numeral buttons of the remote controller to revise the movement speed of the robot, and make different sounds while pressing the remote controller buttons. In the process of programming, try to use the modular building blocks to make the logic of the program clearly.

Program Idea





Reference Code

My Blocks

Make a Block

define Custom block	Define a series of blocks, or a procedure.
Custom block	This block refers to the defined block, or procedure.

The image displays two columns of Scratch code blocks. The left column contains a 'Generate Code' block, a 'set speed to 80' block, a 'forever' loop, and a 'Speed level' block. The 'Speed level' block is a large orange block containing multiple 'if' and 'else' statements that control the speed of a DC motor (M1) based on IR sensor readings (R1-R9) and a limit switch (PortC). The right column contains a 'define Speed level' block, which is a large orange block containing multiple 'if' and 'else' statements that control the speed of a DC motor (M1) based on IR sensor readings (R1-R9) and a limit switch (PortC). The 'define Speed level' block is a large orange block containing multiple 'if' and 'else' statements that control the speed of a DC motor (M1) based on IR sensor readings (R1-R9) and a limit switch (PortC). The 'define Speed level' block is a large orange block containing multiple 'if' and 'else' statements that control the speed of a DC motor (M1) based on IR sensor readings (R1-R9) and a limit switch (PortC).

```

Generate Code
set speed to 80
forever
  Speed level
  set Limit switch detection value to limit switch PortC
  if Limit switch detection value = 1 then
    movement 0 0
  else
    if IR Port2 received ↑ then
      movement speed 0 - speed
    else
      if IR Port2 received ↓ then
        movement 0 - speed speed
      else
        if IR Port2 received ←→ then
          movement speed speed
        else
          if IR Port2 received → then
            movement 0 - speed 0 - speed
          else
            movement 0 0
    if IR Port2 received A then
      dc motor M1 speed -100
    else
      if IR Port2 received D then
        if Limit switch detection value = 1 then
          dc motor M1 speed 0
        else
          dc motor M1 speed 100
      else
        dc motor M1 speed 0

define movement number1 number2
encoder motor Port3 power to number1
encoder motor Port4 power to number2
  
```

```

define Speed level
if IR Port2 received R1 then
  set speed to 40
  play tone on note C4 beat Quarter
else
  if IR Port2 received R2 then
    set speed to 65
    play tone on note D4 beat Quarter
  else
    if IR Port2 received R3 then
      set speed to 90
      play tone on note E4 beat Quarter
    else
      if IR Port2 received R4 then
        set speed to 115
        play tone on note F4 beat Quarter
      else
        if IR Port2 received R5 then
          set speed to 140
          play tone on note G4 beat Quarter
        else
          if IR Port2 received R6 then
            set speed to 165
            play tone on note A4 beat Quarter
          else
            if IR Port2 received R7 then
              set speed to 190
              play tone on note B4 beat Quarter
            else
              if IR Port2 received R8 then
                set speed to 215
                play tone on note C5 beat Quarter
              else
                if IR Port2 received R9 then
                  set speed to 240
                  play tone on note D5 beat Quarter
  
```



Extension: Use the line-following sensor on forklift robot, to realize automatic line-following and objects on the designed route.



G. Camera Dolly Robot

Material

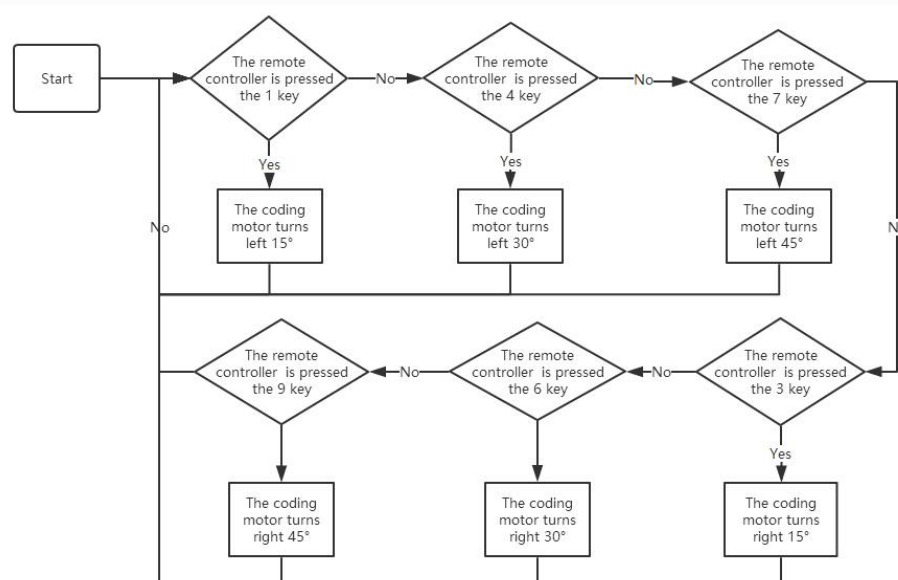
1. Camera dolly robot
2. Computer



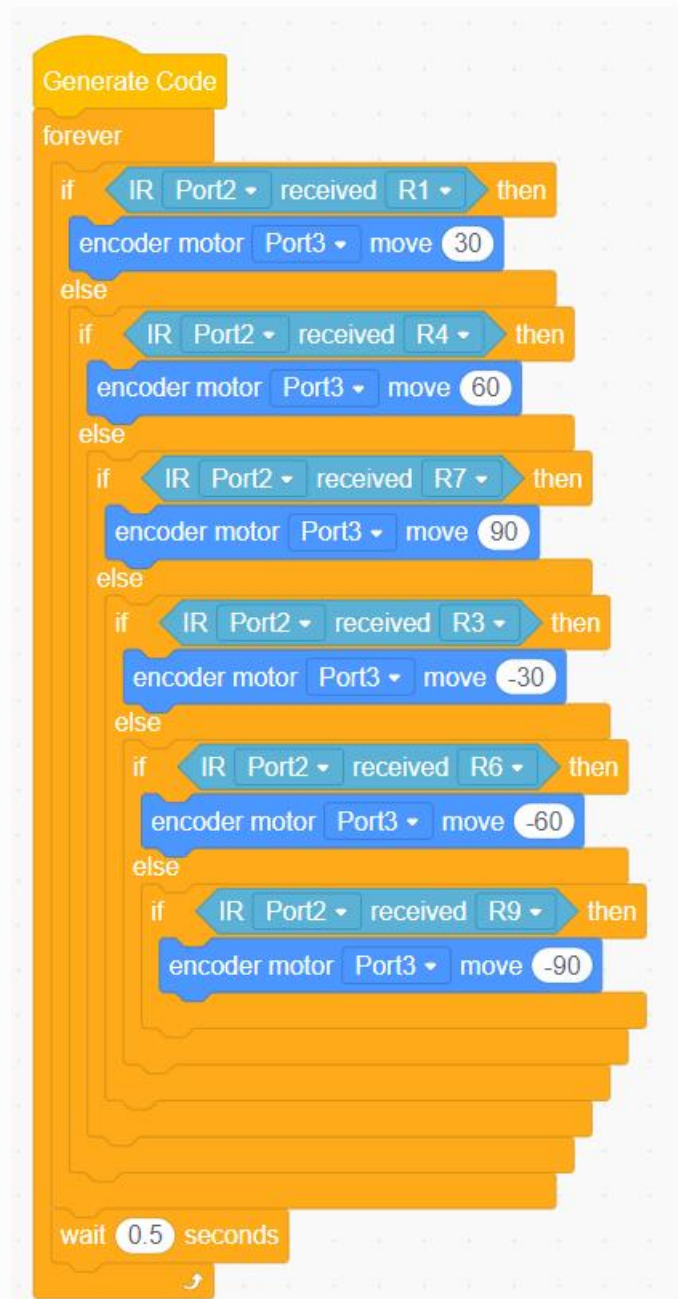
Project #1: Accurate Control of Encoder DC Motor

Use the block "set encoder motor power to..." to precisely control the turning angle of the wheel which is in front of the robot.

Program Idea



Reference Code

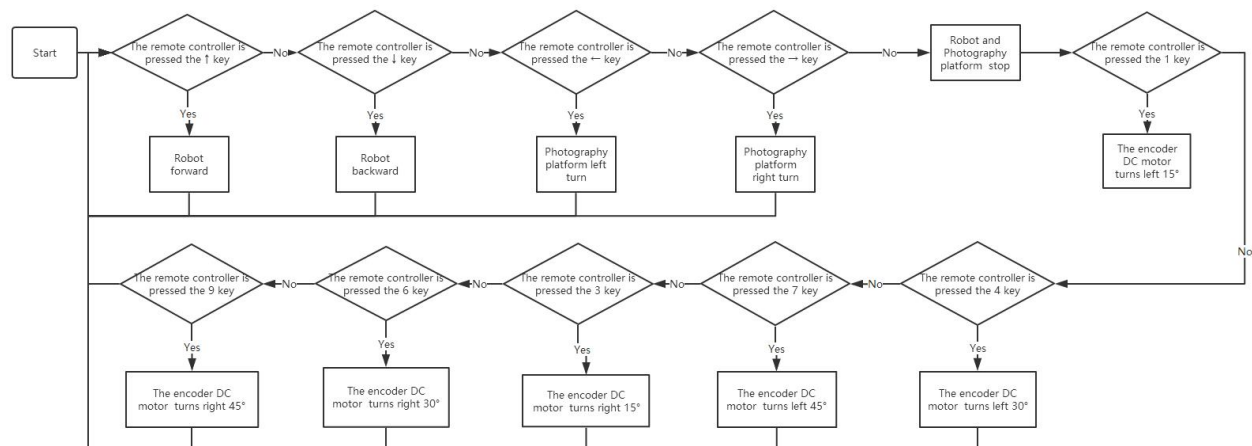


Note: When use the block "set encoder motor power to..." in power supply state, if there is an external force trying to roll the wheel, the encoder will rotate in the opposite direction to keep the wheel at the original angle.

Project #2: Remote Control Photography

Use the up and down buttons of the remote controller to control the robot's forward and backward movement. Use the left and right buttons of the remote controller to control the left and right turns of the camera platform. Use the button 1, 4 and 7 of the remote control to control the robot's left turns at different angles. Use the button 3, 6 and 9 of the remote control to control the robot's right turns at different angles. Place the mobile phone on the photographic platform to transit the images captured from one mobile phone to another mobile phone through WeChat or video capture software to realize remote transmission of images.

Program Idea



Reference Code



Expansion: Try to install the camera structure on the forklift robot and remotely control the forklift robot to carry objects through the camera.

H. 4DOF Robot Arm

Material

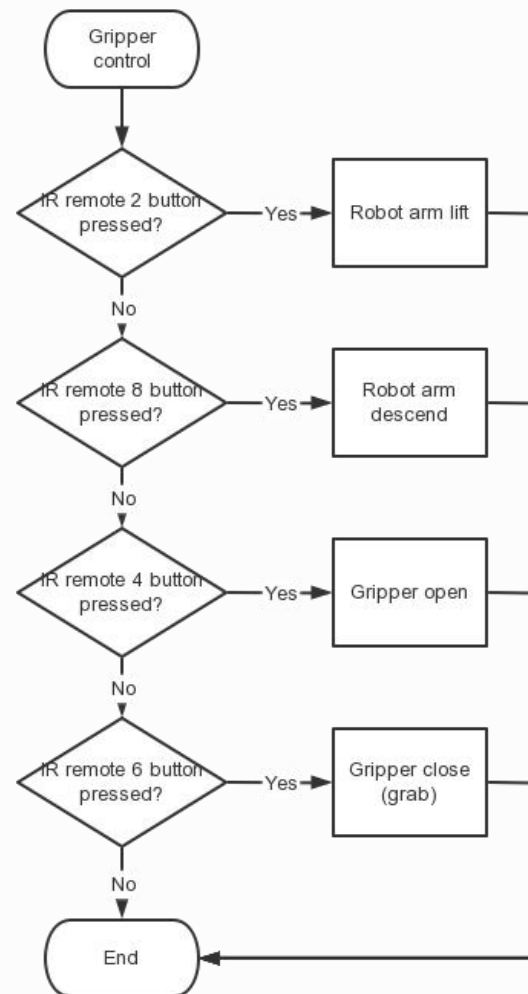
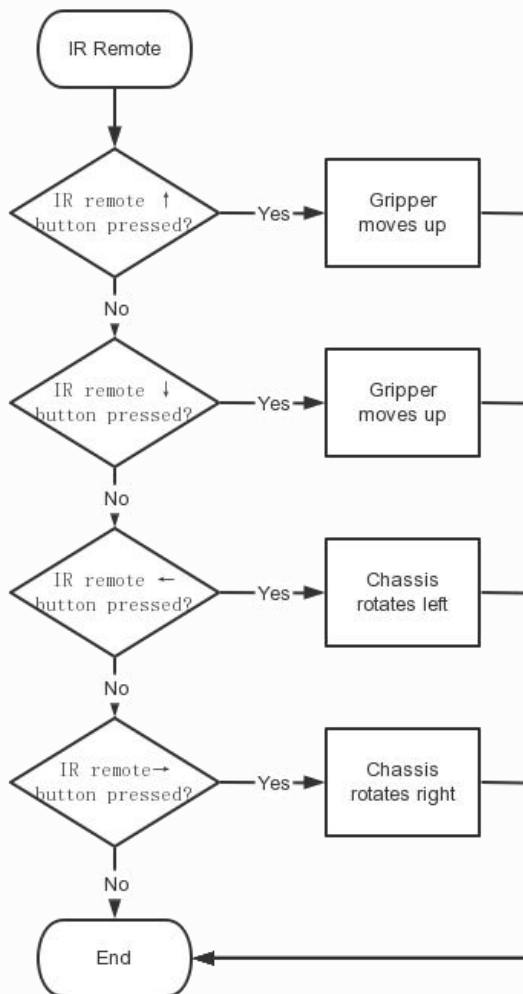
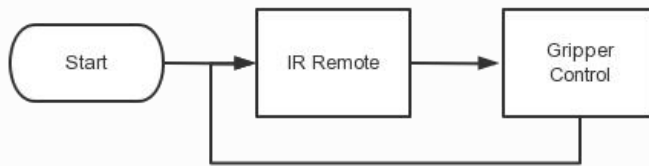
1. 4DOF Robot Arm
2. Computer



Project #1: Remote control three-axis mechanical arm

1. Utilize the remote controller's up and down buttons to regulate the M1 motor, which governs the vertical motion of the first mechanical arm.
2. Employ the remote controller's left and right buttons to command the DC motor in Port 3, which determines the lateral movement of the robot arm.
3. Utilize the remote controller's 2 and 8 buttons to operate the DC motor in Port 4, thus controlling the vertical swing of the second mechanical arm.
4. Utilize the remote controller's 4 and 6 buttons to manipulate the Port B servo motor, responsible for the actuation of the robot gripper's opening and closing mechanism.

Program Idea



Reference Code

The image displays a Scratch script for controlling a robotic gripper. The script is organized into three main sections: a main loop, an IR remote control function, and a gripper control function.

Main Loop:

- Generate Code** block.
- set servo motor angle** to 0.
- forever** loop containing:
 - IR remote** block.
 - Gripper control** block.

define IR remote:

- if** IR Port2 received **↑** **then**: **dc motor M1** speed 100.
- else**:
 - if** IR Port2 received **↓** **then**: **dc motor M1** speed -100.
 - else**:
 - if** IR Port2 received **←** **then**: **encoder motor Port3** power to 100.
 - else**:
 - if** IR Port2 received **→** **then**: **encoder motor Port3** power to -100.
 - else**: **dc motor M1** speed 0; **encoder motor Port3** power to 0.

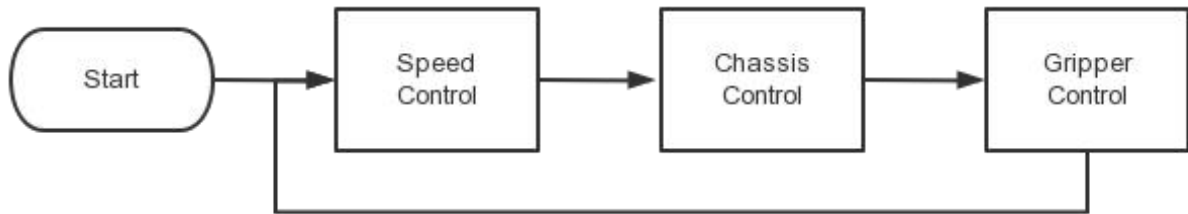
define Gripper control:

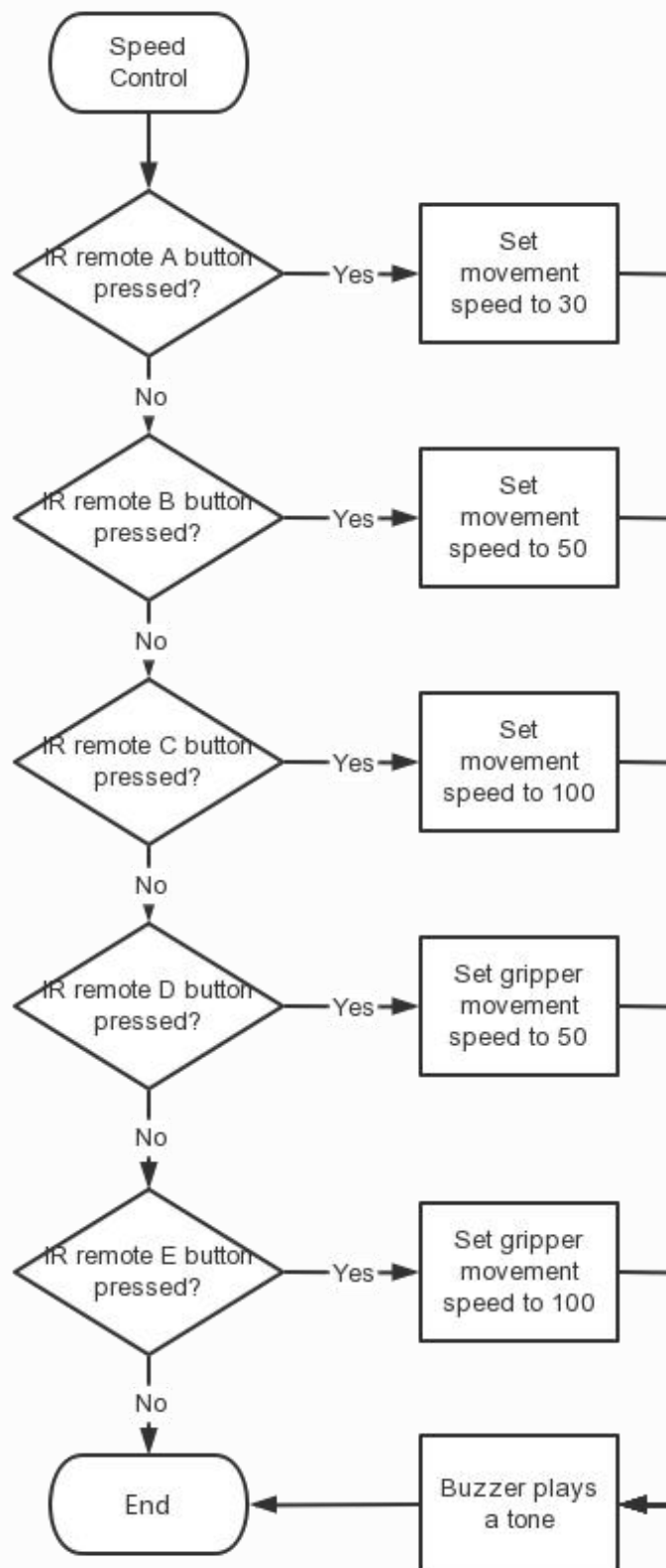
- if** IR Port2 received **R2** **then**: **encoder motor Port4** power to 100. (Linked to **Arm lift** block).
- else**:
 - if** IR Port2 received **R8** **then**: **encoder motor Port4** power to -100. (Linked to **Arm descend** block).
 - else**:
 - if** IR Port2 received **R4** **then**:
 - if** **servo motor angle < 80** **then**: **change servo motor angle** by 1. (Linked to **Gripper open...** block).
 - else**:
 - if** IR Port2 received **R6** **then**:
 - if** **0 < servo motor angle** **then**: **change servo motor angle** by -1. (Linked to **Gripper clos...** block).
 - else**: **encoder motor Port4** power to 0.
- wait 0.01 seconds**.
- servo PortD angle** servo motor angle.

Project #2: Creating a variable speed three-axis robotic arm.

Building upon project 1, an enhanced feature is introduced to the remote-controlled 4DOF robot arm, allowing for speed regulation. The A, B, and C buttons on the remote controller are utilized to adjust the lateral movement speed of the chassis, while the D and E buttons are used to modify the speed of the mechanical arm (near the mechanical claw).

Program Idea





Reference Code

```

Generate Code
set servo motor angle to 0
set speed to 100
set robot gripper movement speed to 100
forever
  chassis control
  Speed control
  robot gripper control

```

```

define Speed control
if IR Port2 received A then
  set speed to 30
  play tone on note C4 beat Half
else
  if IR Port2 received B then
    set speed to 50
    play tone on note C4 beat Half
  else
    if IR Port2 received C then
      set speed to 100
      play tone on note C4 beat Half
    else
      if IR Port2 received D then
        set robot gripper movement speed to 50
        play tone on note C4 beat Half
      else
        if IR Port2 received E then
          set robot gripper movement speed to 100
          play tone on note C4 beat Half

```

Extension: You can place a lot of small objects around the three-axis manipulator, and control the manipulator to move them to the same place or stack them up in a certain period of time.

```

define robot gripper control
if IR Port2 received R2 then
  gripper lift
  encoder motor Port4 power to robot gripper movement speed
else
  if IR Port2 received R8 then
    gripper desc...
    encoder motor Port4 power to 0 - robot gripper movement speed
  else
    if IR Port2 received R4 then
      gripper open
      if servo motor angle < 80 then
        change servo motor angle by 1
    else
      if IR Port2 received R6 then
        gripper clos...
        if 0 < servo motor angle then
          change servo motor angle by -1
        else
          encoder motor Port4 power to 0
  wait 0.01 seconds
  servo PortD angle servo motor angle

```

```

define chassis control
if IR Port2 received ↑ then
  dc motor M1 speed 100
else
  if IR Port2 received ↓ then
    dc motor M1 speed -100
  else
    if IR Port2 received ← then
      encoder motor Port3 power to speed
    else
      if IR Port2 received → then
        encoder motor Port3 power to 0 - speed
    else
      dc motor M1 speed 0
      encoder motor Port3 power to 0

```

I. Bartender Robot

Material

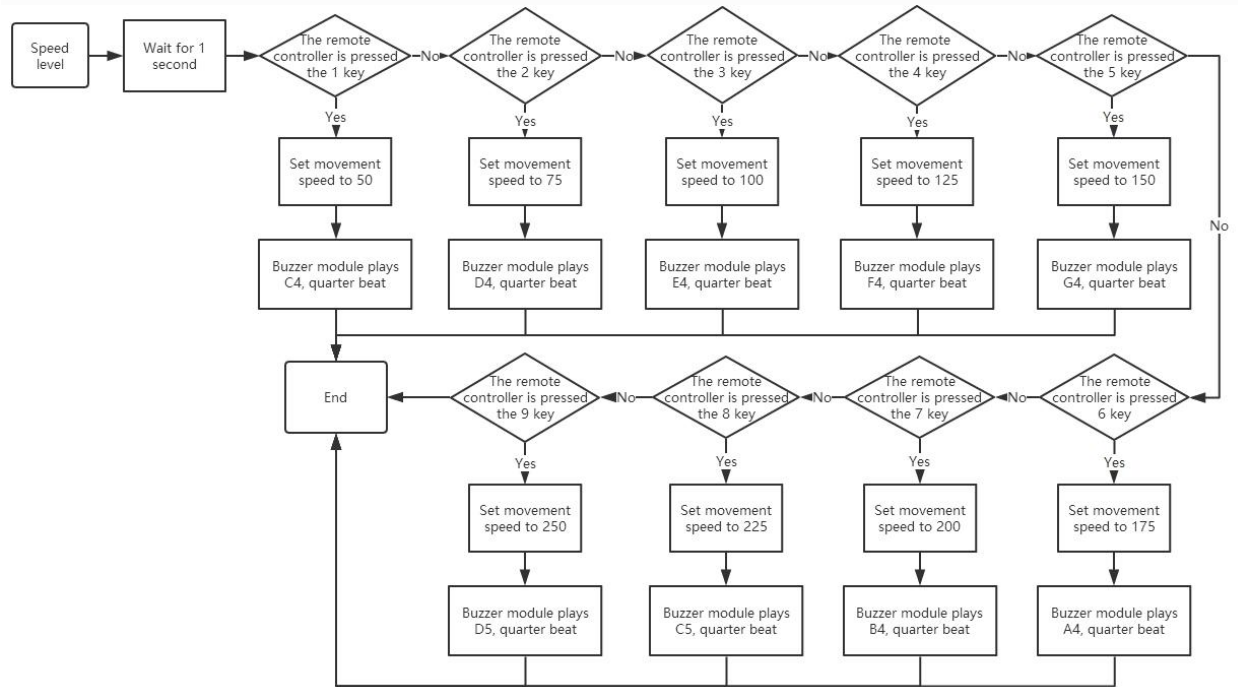
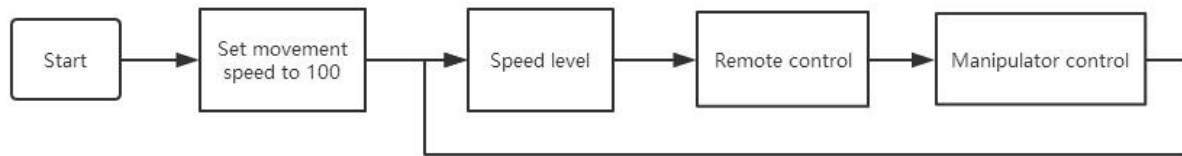
1. Bartender Robot
2. Computer

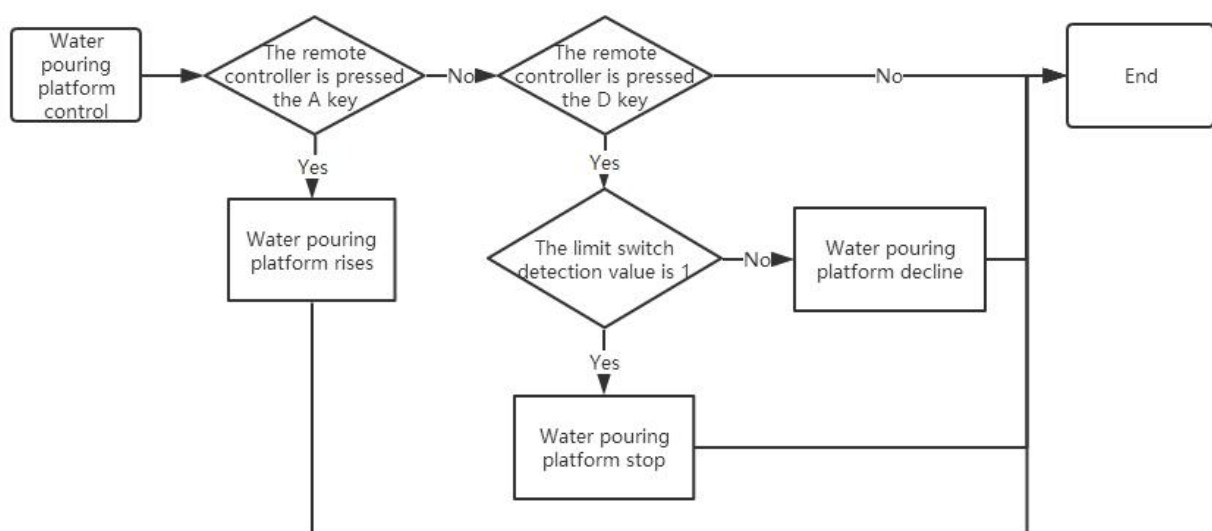
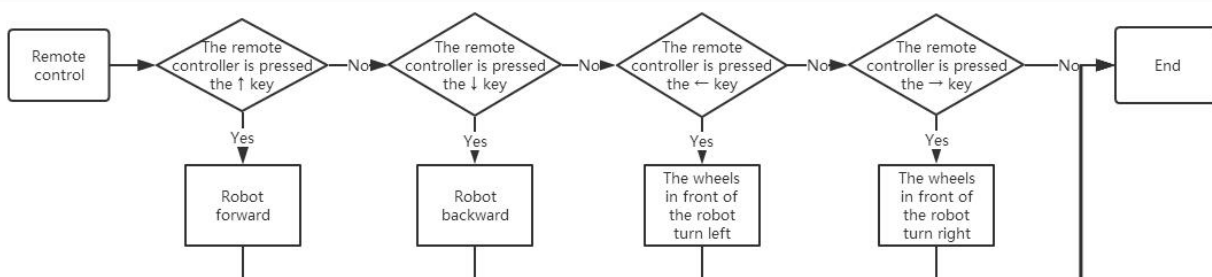


Project #1: Remote control Bartender Robot

The up and down buttons of the remote controller are used to control the forward and backward movement of the robot, the left and right buttons of the remote controller are used to control the left and right turn of the wheels in front of the robot, and the A and D buttons of the remote controller is used to control the up and down tilt of the water pouring platform to pour water. It should be noted here that the limit switch module is installed on the water pouring platform. Therefore, if the water pouring platform rises too high, the limit switch will be triggered, at this time, pressing the button of the remote control will cause the robot will not respond. At the same time, we can use the 1 to 9 buttons of the remote controller to control the speed of the robot, and the buzzer will make different sounds when a different button is pressed, so that we can distinguish the pressed buttons.

Program Idea





Reference Code

The image displays a collection of Scratch code blocks organized into four main sections:

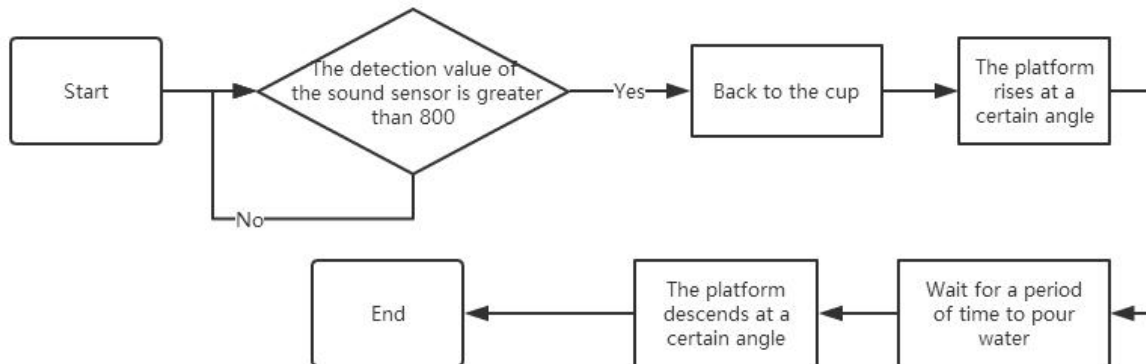
- Generate Code:** A block that sets the speed to 100 and enters a 'forever' loop containing three sub-functions: 'Speed Level', 'Remote Control', and 'Control the Water Pouring Platform'.
- define: Remote Control:** A function that checks for IR signals on Port2. It uses a series of 'if-then' and 'else' blocks to set the power of encoder motors Port3 and Port4 based on the received signal (up, down, left, right, or no signal).
- define: Speed Level:** A function that checks for IR signals on Port2 (R1 through R9). It uses a series of 'if-then' and 'else' blocks to set the speed of a DC motor (M1) and play a specific note on a piano (C4, D4, E4, F4, G4, A4, B4, C5, D5) based on the received signal.
- define: Control the Water Pouring Platform:** A function that checks for a limit switch on PortC and an IR signal on Port2 (A). It uses 'if-then' and 'else' blocks to set the speed of a DC motor (M1) to 150, 0, or -150 based on the received signal.

The code is written in a visual programming style using colored blocks (orange for loops, blue for logic, purple for sound, and green for variables) on a grid background.

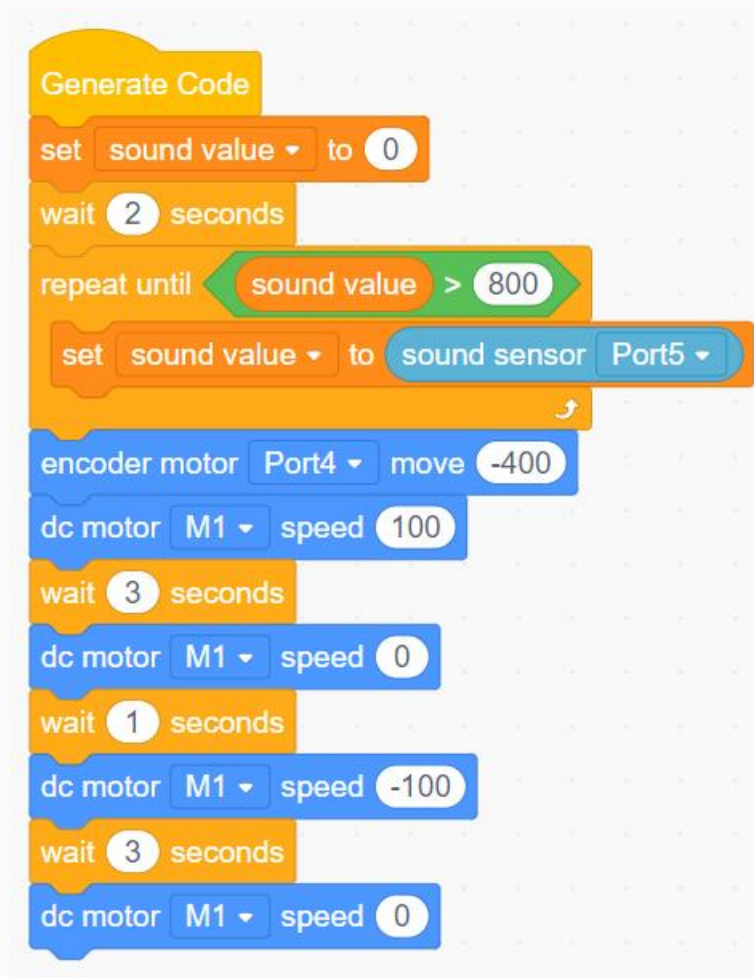
Project #2: Automatic Bartender Robot

Place the water cup and robot in position ahead of time, and then the forward distance of the robot is adjusted according to the positional relationship between the two, so that it just moves to the side of the water cup, next, the fall time and the pouring time are adjusted according to the position of the robot's pouring platform. In addition, we can use clapping as its starting condition to realize the function of automatically pouring water as soon as clapping.

Program Idea



Reference Code



Note: This program is only a reference program. The parameters in the program should be adjusted according to the actual situation.

J. Robot Arm Tank

Material

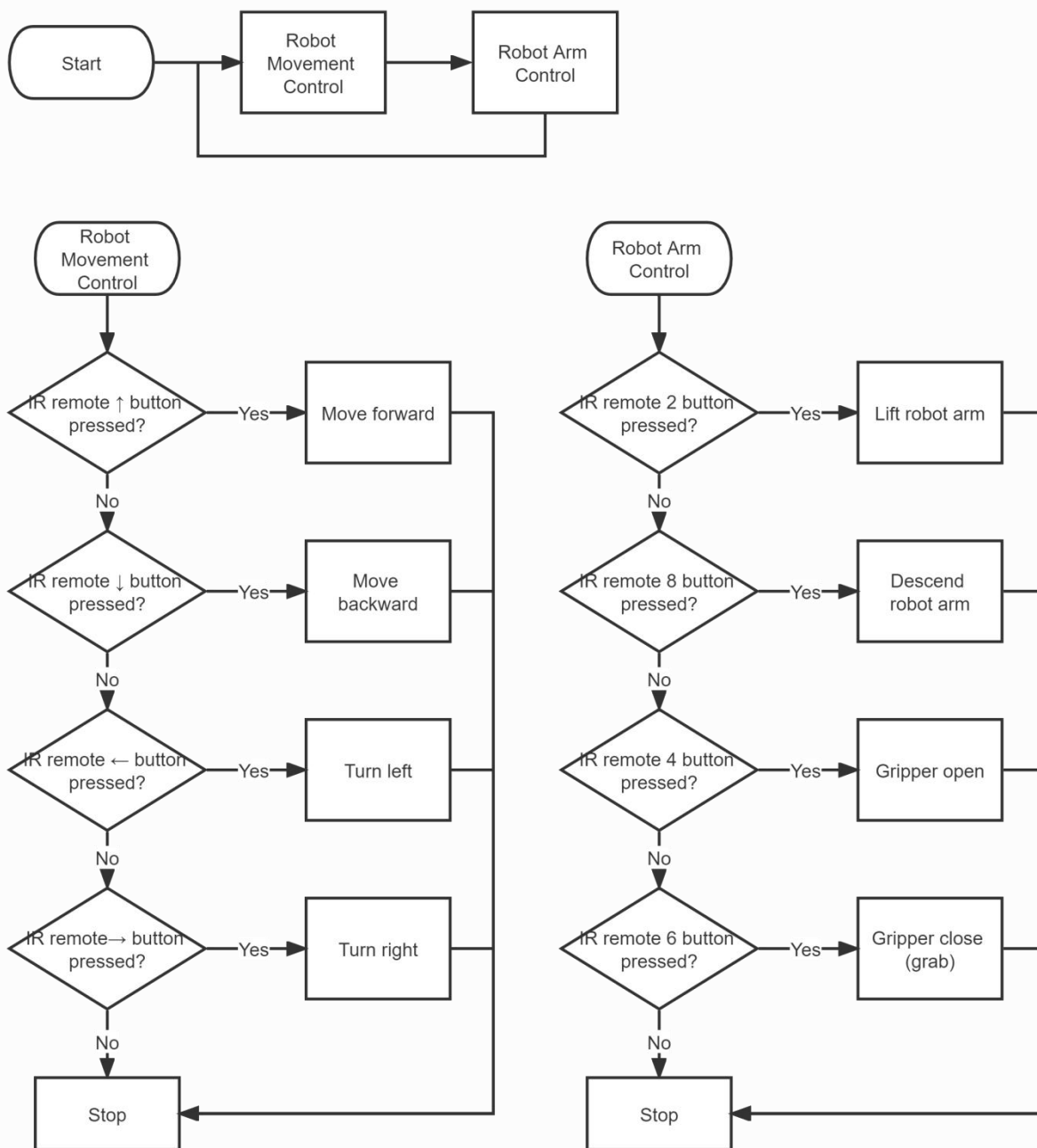
1. Robot Arm Tank
2. Computer



Project #1: Create a remote-controlled robotic arm tank

Utilize the IR remote controller's up, down, left, and right buttons to govern the robot's movement. Deploy button 2 and button 8 of the remote controller to regulate the ascent and descent of the robot arm, and utilize button 4 and button 6 to operate the opening and closing of the robot gripper claw.

Program Idea



Reference Code

The image displays two custom code blocks for a robot, 'robot movement control' and 'robot arm control', along with a main sequence block.

Main Sequence:

- Generate Code
- set servo motor angle to 0
- forever loop:
 - robot movement control
 - robot arm control

define robot movement control

- if IR Port2 received ↑ then
 - encoder motor Port3 power to 100
 - encoder motor Port4 power to -100
- else
 - if IR Port2 received ↓ then
 - encoder motor Port3 power to -100
 - encoder motor Port4 power to 100
 - else
 - if IR Port2 received ← then
 - encoder motor Port3 power to 100
 - encoder motor Port4 power to 100
 - else
 - if IR Port2 received → then
 - encoder motor Port3 power to -100
 - encoder motor Port4 power to -100
 - else
 - encoder motor Port3 power to 0
 - encoder motor Port4 power to 0

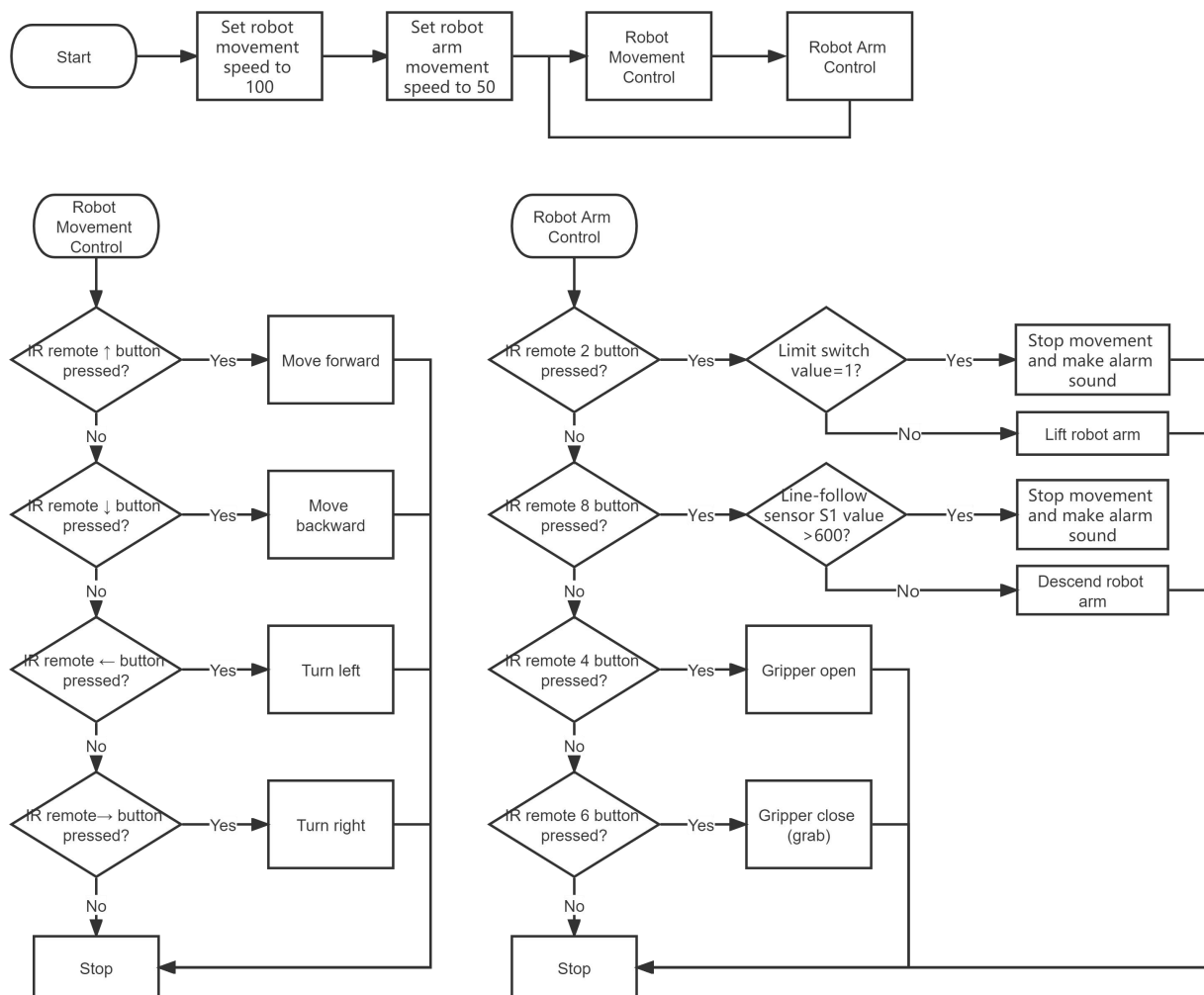
define robot arm control

- if IR Port2 received R2 then
 - dc motor M1 speed -50
- else
 - if IR Port2 received R8 then
 - dc motor M1 speed 50
 - else
 - if IR Port2 received R4 then
 - if servo motor angle < 180 then
 - change servo motor angle by 1
 - else
 - if IR Port2 received R6 then
 - if 0 < servo motor angle then
 - change servo motor angle by -1
 - else
 - dc motor M1 speed 0
- wait 0.01 seconds
- servo PortD angle servo motor angle

Project #2: Remote control posture limitation for the robotic arm tank

On the basis of project 1, we add a limit switch module and a double-way line-following module, which both are used to limit the movement range of the robot arm to avoid damaging the mechanical structure.

Program Idea



Reference Code

```

Generate Code
set servo motor angle to 0
set speed to 100
set gripper movement speed to 50
forever
  control robot movement
  control robot arm

```



```

define control robot movement
if IR Port2 received ↑ then
  encoder motor Port3 power to speed
  encoder motor Port4 power to 0 - speed
else
  if IR Port2 received ↓ then
    encoder motor Port3 power to 0 - speed
    encoder motor Port4 power to speed
  else
    if IR Port2 received ← then
      encoder motor Port3 power to speed
      encoder motor Port4 power to speed
    else
      if IR Port2 received → then
        encoder motor Port3 power to 0 - speed
        encoder motor Port4 power to 0 - speed
      else
        encoder motor Port3 power to 0
        encoder motor Port4 power to 0

```

```

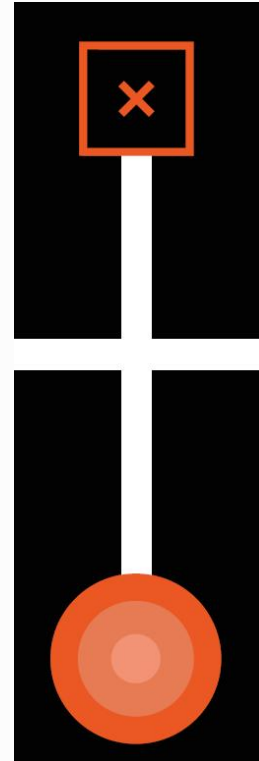
define control robot arm
set limit switch value to limit switch PortC
set s1 to line follower PortB S1
if IR Port2 received R2 then
  if limit switch value = 1 then
    dc motor M1 speed 0
    play tone on note C4 beat Half
  else
    dc motor M1 speed 0 - gripper movement speed
  else
    if IR Port2 received R8 then
      if s1 > 600 then
        dc motor M1 speed 0
        play tone on note C4 beat Half
      else
        dc motor M1 speed gripper movement speed
    else
      if IR Port2 received R4 then
        if servo motor angle < 180 then
          change servo motor angle by 1
        else
          if IR Port2 received R6 then
            if 0 < servo motor angle then
              change servo motor angle by -1
            else
              dc motor M1 speed 0
          else
            wait 0.05 seconds
            servo PortD angle servo motor angle

```

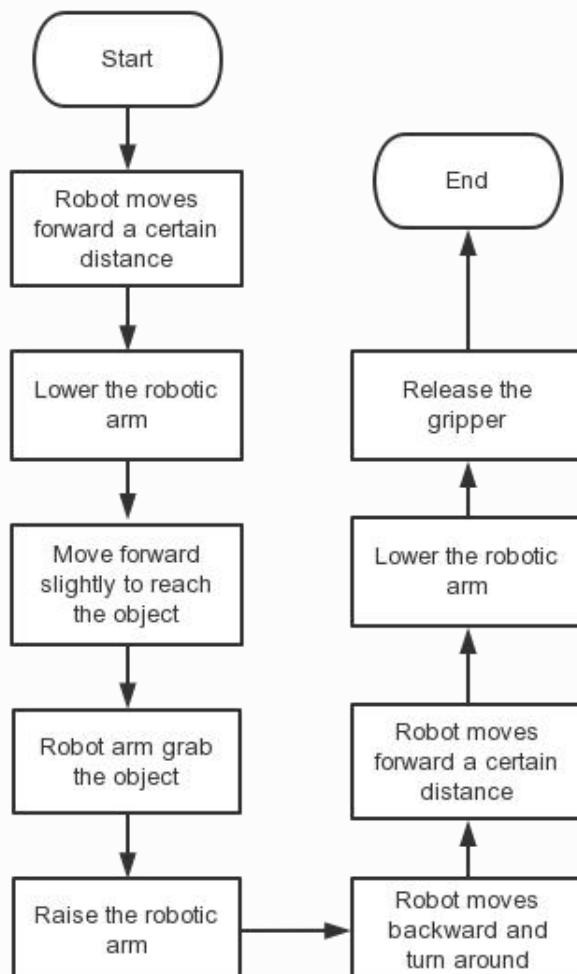
Project #3: Create a fully automated transportation robot

Design a route for an automated transportation robot.

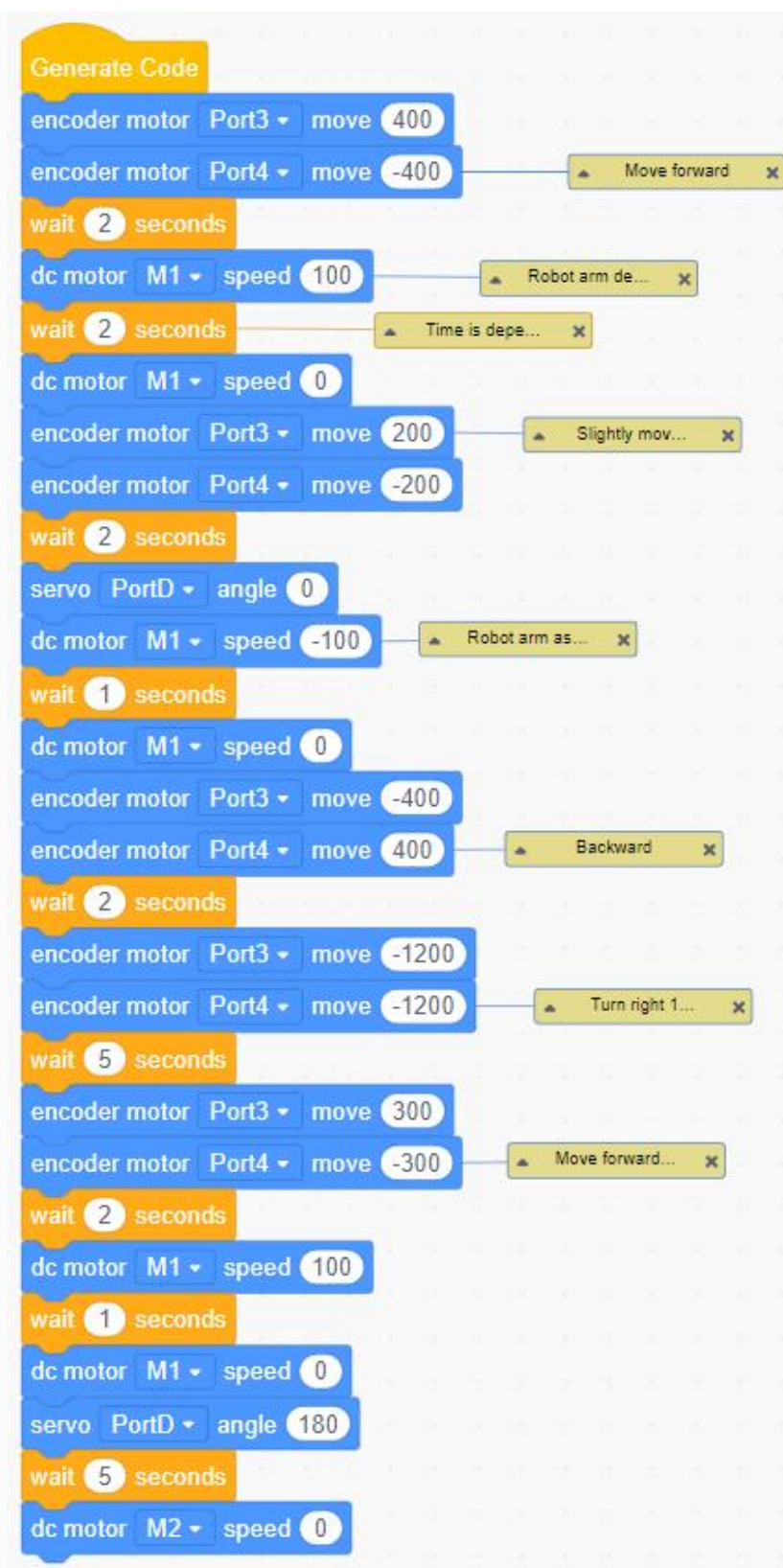
Assuming that the object blocks are located within a circular ring and the robot is positioned at the intersection, the task is to transport the object blocks into the square box. Program the robot to automatically perform the task.



Program Idea



Reference Code





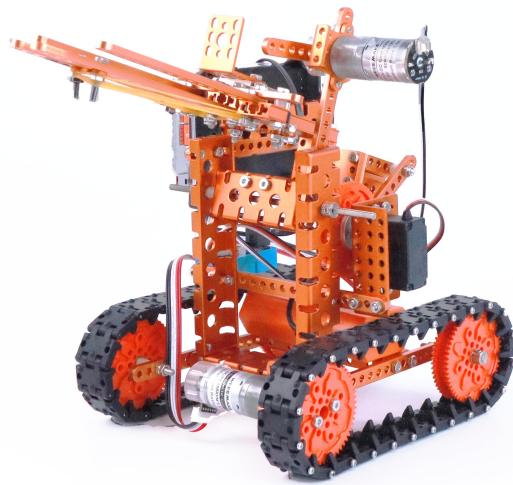
Note: This program is only a reference program. The parameters in the program should be adjusted according to the actual situation.

Expansion: In this lesson, we learned how to control the robotic arm tank robot. Can we combine it with a camera robot to create a robot that integrates photography function and robotic arm? Or can we combine it with a three-axis robotic arm to have two robots working together for material handling, achieving a conveyor belt-like object transportation system similar to a factory?

K. Ball Launcher

Material

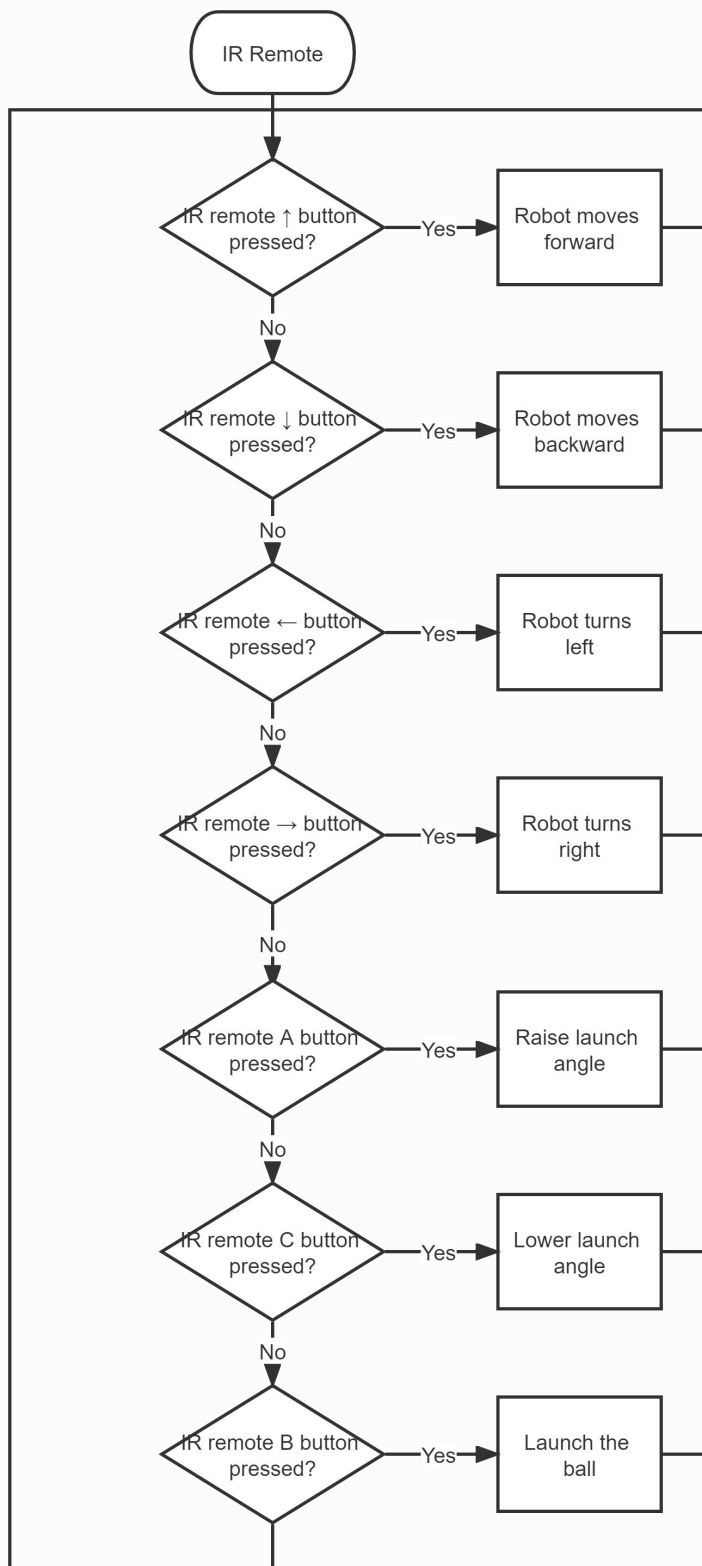
1. Ball Launcher Robot
2. Computer



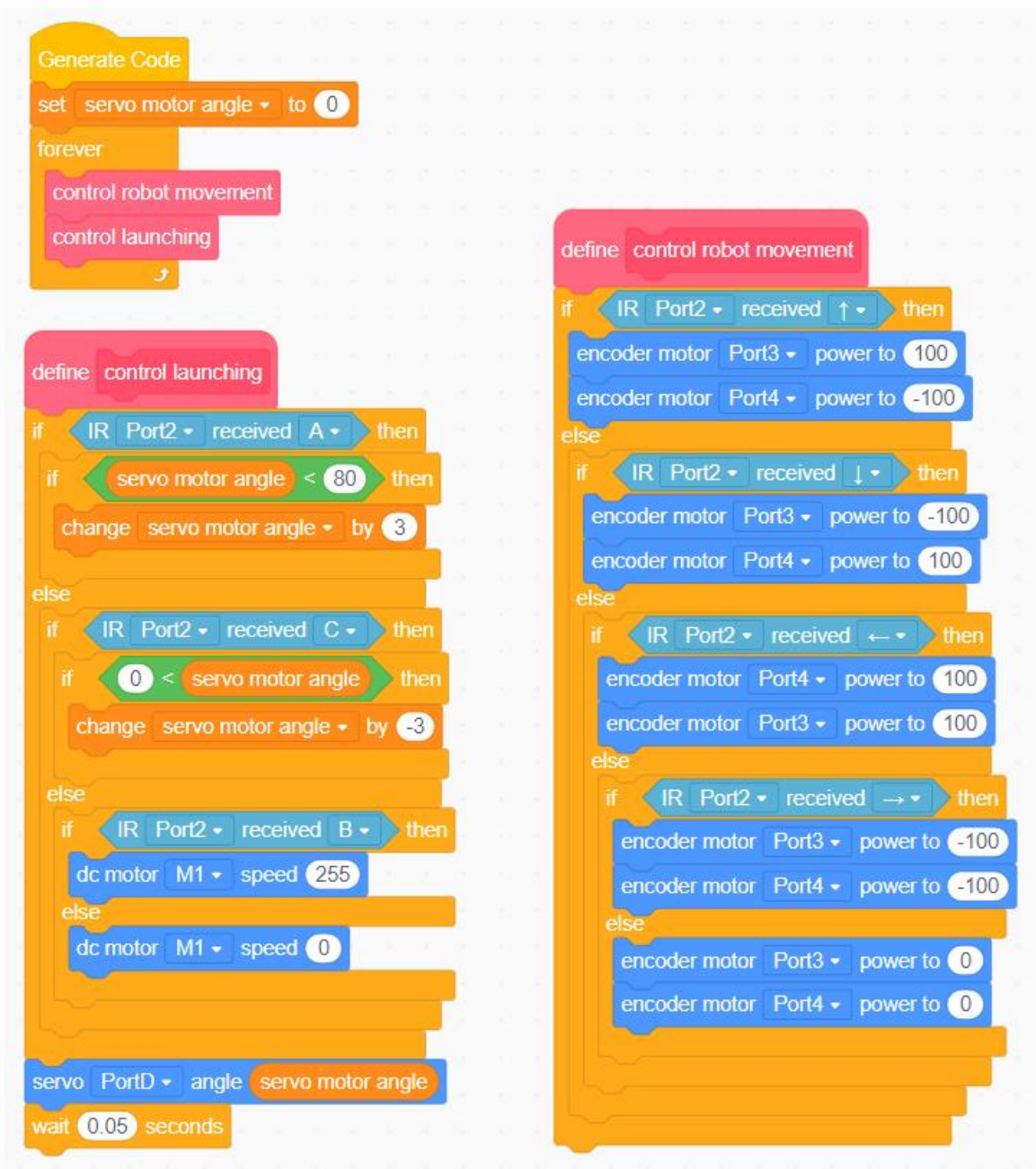
Project #1: Create a remote-controlled ball-launching robot.

In this educational activity, we will learn to operate the robot chassis using the up, down, left, and right buttons on the remote controller. We will also discover how to control the shooting platform's elevation by using the A and C buttons. Finally, we will have the opportunity to showcase their skills by initiating a launch with the press of the B button.

Program Idea



Reference Code

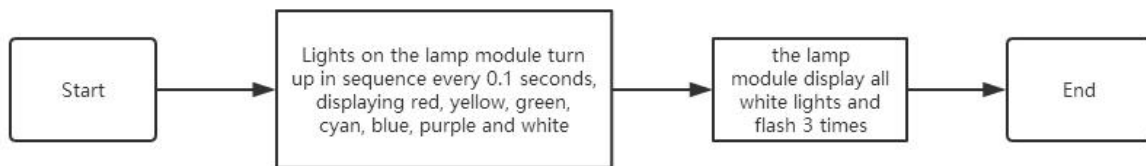


Note: The 3D printing files for the ball launcher's storage box can be downloaded from the official website.

Project #2: Add launch prompt function

Utilize the RGB LED-8 module plugged on the mainboard to create a captivating visual display. The module will showcase a sequence of vibrant colors, including red, orange, yellow, green, cyan, blue, purple, and white. As a signal prompt preceding the launch of the projectile, All 8 RGB lights will flash three times in white in a visually striking pattern.

Program Idea



Reference Code

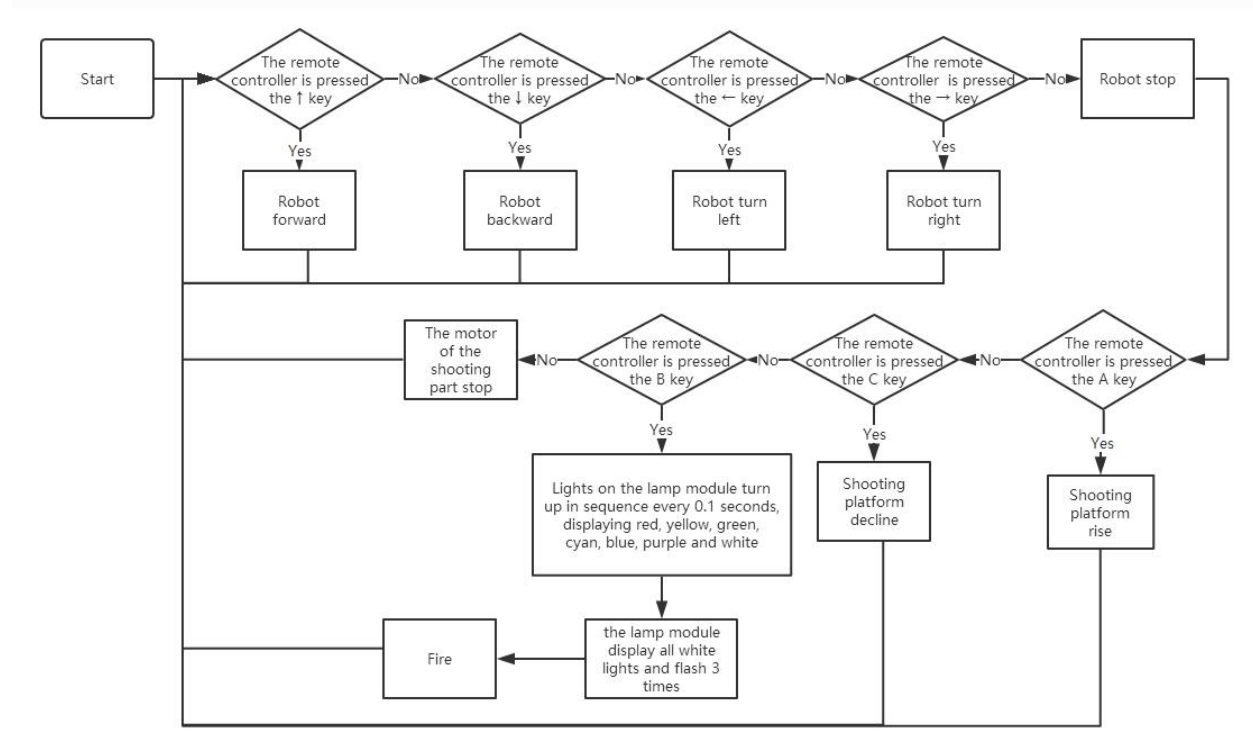
```

Generate Code
rgb ring 1 red 255 green 0 blue 0
wait 0.1 seconds
rgb ring 2 red 255 green 150 blue 0
wait 0.1 seconds
rgb ring 3 red 255 green 255 blue 0
wait 0.1 seconds
rgb ring 4 red 0 green 255 blue 0
wait 0.1 seconds
rgb ring 5 red 0 green 150 blue 150
wait 0.1 seconds
rgb ring 6 red 0 green 0 blue 255
wait 0.1 seconds
rgb ring 7 red 0 green 0 blue 255
wait 0.1 seconds
rgb ring 8 red 255 green 150 blue 255
wait 0.1 seconds
repeat 3
  rgb ring 0 red 255 green 255 blue 255
  wait 0.2 seconds
  rgb ring 0 red 0 green 0 blue 0
  wait 0.2 seconds
  
```

Project #3: Optimize the program of the remote-controlled ball-launching robot

By combining project 1 and 2, when the B button on the remote controller is pressed, the robot will initiate a light signal prompt before launching. Once the prompt is complete, the robot will proceed with the launching action.

Program Idea



Reference Code

```

Generate Code
set servo motor angle to 0
forever
  control robot movement
  control launching

```

```

define control robot movement
if IR Port2 received ↑ then
  encoder motor Port3 power to 100
  encoder motor Port4 power to -100
else
  if IR Port2 received ↓ then
    encoder motor Port3 power to -100
    encoder motor Port4 power to 100
  else
    if IR Port2 received ← then
      encoder motor Port4 power to 100
      encoder motor Port3 power to 100
    else
      if IR Port2 received → then
        encoder motor Port3 power to -100
        encoder motor Port4 power to -100
      else
        encoder motor Port3 power to 0
        encoder motor Port4 power to 0

```

```

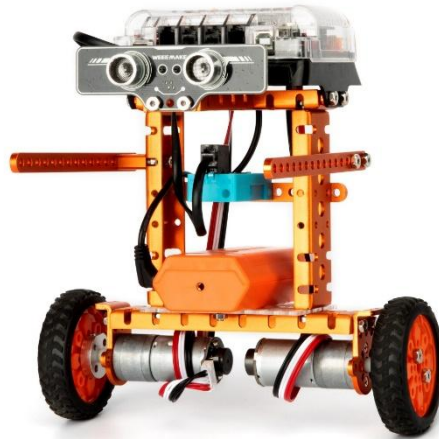
define control launching
if IR Port2 received A then
  if servo motor angle < 80 then
    change servo motor angle by 2
  else
    if IR Port2 received C then
      if 0 < servo motor angle then
        change servo motor angle by -2
    else
      if IR Port2 received B then
        rgb ring 1 red 255 green 0 blue 0
        wait 0.1 seconds
        rgb ring 2 red 255 green 150 blue 0
        wait 0.1 seconds
        rgb ring 3 red 255 green 255 blue 0
        wait 0.1 seconds
        rgb ring 4 red 0 green 255 blue 0
        wait 0.1 seconds
        rgb ring 5 red 0 green 150 blue 150
        wait 0.1 seconds
        rgb ring 6 red 0 green 0 blue 255
        wait 0.1 seconds
        rgb ring 7 red 255 green 150 blue 255
        wait 0.1 seconds
        rgb ring 8 red 255 green 255 blue 255
        wait 0.1 seconds
        repeat 3
          rgb ring 0 red 255 green 255 blue 255
          wait 0.2 seconds
          rgb ring 0 red 0 green 0 blue 0
          wait 0.2 seconds
        ,
        dc motor M1 speed 255
        wait 0.8 seconds
        dc motor M1 speed 0
        wait 1 seconds
      else
        dc motor M1 speed 0
    ,
    servo PortD angle servo motor angle
    wait 0.05 seconds

```

L. Self-balancing Robot

Material

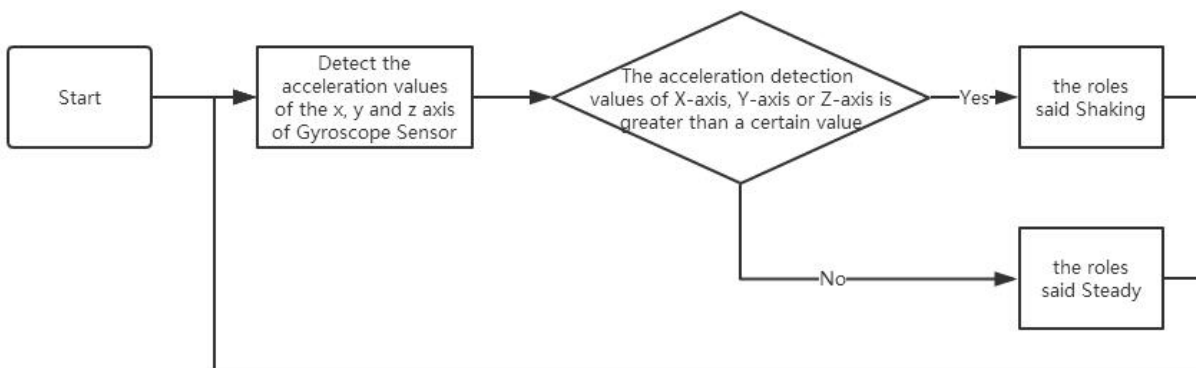
1. Self-balancing robot
2. Computer



Project #1: Mastering the use of gyroscope sensor

Use the block " Gyro sensor... acceleration..." to detect the accelerations of the X axis, Y axis and Z axis of the gyroscope, so as to learn changes of the acceleration of Gyroscope Sensor, and combine the roles with it. When the gyroscope is stationary or shaking, the roles can perform corresponding actions.

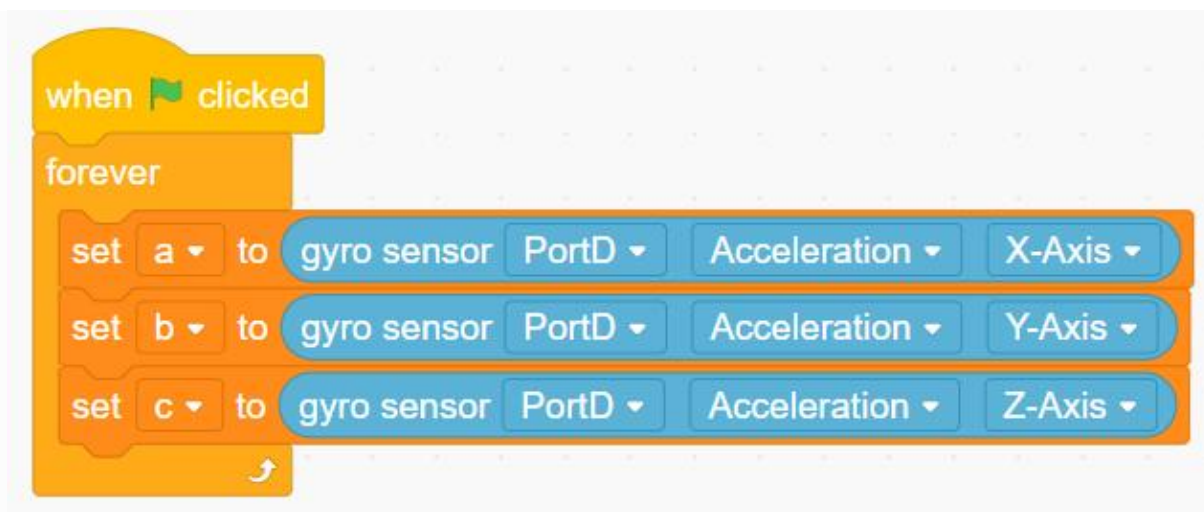
Program Idea



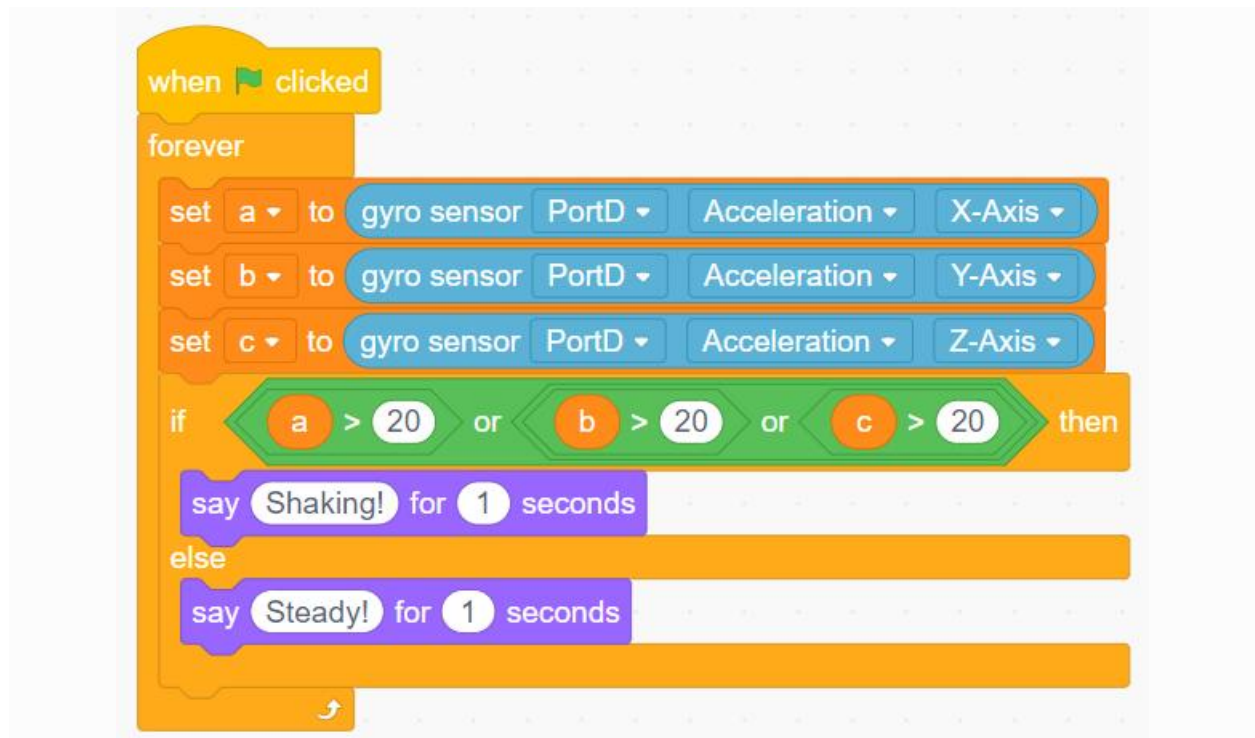
Reference Code

gyro sensor PortA ▾ Gyration ▾ X-Axis ▾	Obtaining the rotation angle/rotational acceleration value on a certain axis.
---	---

Acceleration Measurement of Gyro Sensor



Application of Gyro Sensor Acceleration



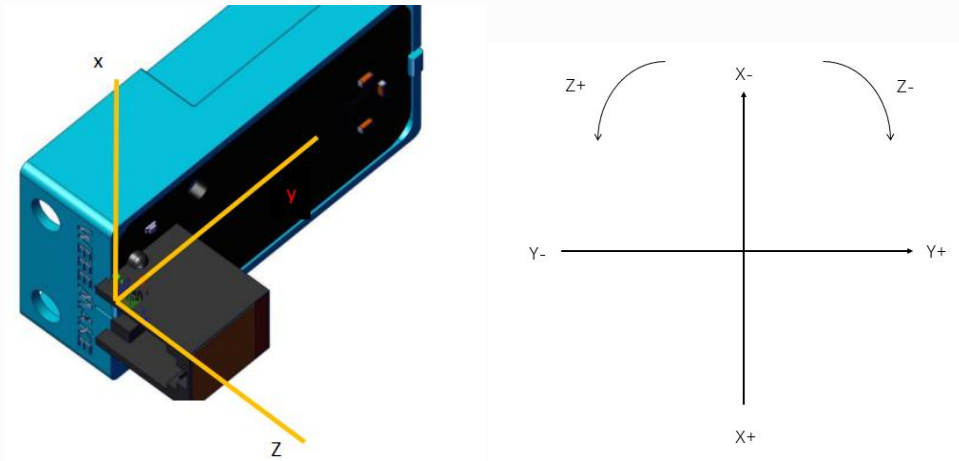
Note: To detect the acceleration value of the axis, it is best to disassemble the sensor from the robot.

After running the program, when the gyroscope is stationary, the roles will say "Steady!" for 1 second, when the gyroscope moves, the roles will say "Shaking!" for 1 second.

Project #2: Testing and application programs for gyroscope sensor

Use the block " Gyro sensor ..." to detect the angles of the X axis, Y axis and Z axis of the gyroscope, so as to learn the changes of the gyroscope sensor angle, and combine the roles with it. When the gyroscope tilts or rotates, the roles can perform corresponding actions.

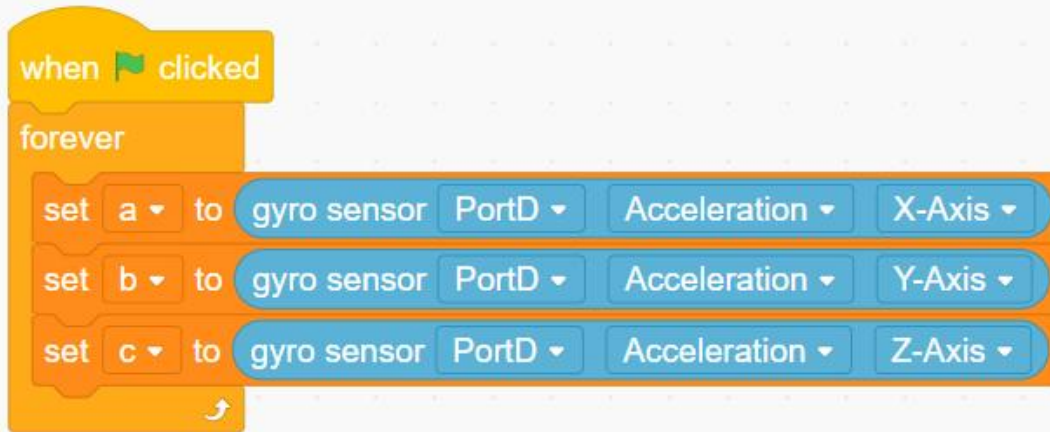
Program Idea



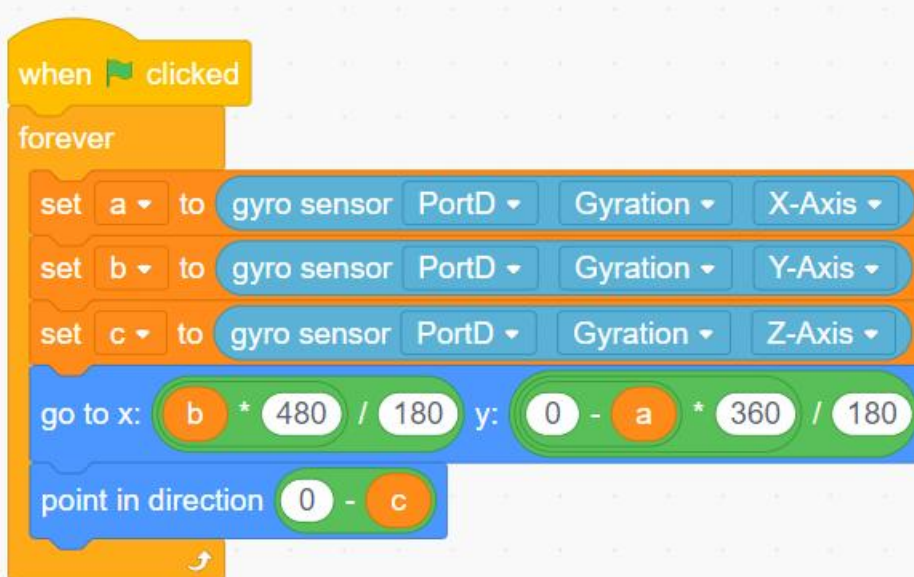
Note: In the context of the gyroscope sensor, **acceleration** refers to the change in speed that occurs when it rotates around an axis. We can think of it as rotational acceleration. Similarly, when the gyroscope rotates around an axis, it also experiences a change in angle. So, we can say that it measures the rotational angle.

Reference Code

Angle Measurement of Gyro Sensor



Application of Gyro Sensor Angle



After executing the program, when the gyroscope leans forward, the role moves upward. When the gyroscope leans backward, the role moves down. When the gyroscope leans to the left, the role moves to the left. When the gyroscope moves to the right, the role moves to the right.

Project #3: Experience the self-balancing robot.

It is difficult to write the program for a self-balancing robot by graphical programming as it requires a very complicated processing of the data detected by gyroscope. So in here we just observe how the self-balancing robot works.

We should restore the factory program onto the mainboard firstly, select the 12-in-1 storm in the factory firmware. Then we make the robot upright, turn on the power switch, and press the D key of the remote controller to enter the self-balancing mode. We can learn the self-balancing robot keeps walking upright. At the same time, other robots we made before also can realize rapid experience by restoring the factory firmware.

Button A——Manual Mode

Button B——Obstacle Avoidance

Button C——Line-following Mode

Button D——Self-balancing Mode

Button E——Robot Arm Mode

(In this mode:

Button ←——Port 3 Encoder Motor A clockwise

Button →——Port 3 Encoder Motor A anticlockwise

Button ↑——Port 4 Encoder Motor B clockwise

Button ↓——Port 4 Encoder Motor B anticlockwise)

Button F——RGB LED Circle

(In this mode:

Button 1——Rotating water light;

Button 2——Rainbow light;

Button 3——Turn off the light)

Button ↑——Forward

Button ↓——Backward

Button ←——Turn left

Button →——Turn right

Button OK——RGB on ultrasonic sensor

Number Button 0——Speed 0

Number Button 1——Speed 1

Number Button 2——Speed 2

Number Button 3——Speed 3

Number Button 4——Speed 4

Number Button 5——Speed 5

Number Button 6——M1 clockwise

Number Button 7——Port D/M2 gripper clamp

Number Button 8——Port D/M2 gripper release

Number Button 9——M1 anti-clockwise